

Kandidatuppsats

Statistiska Institutionen

Nr 2026:x

Title in English

Title in Swedish

May 14, 2026

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for the Alzheimer's Disease Neuroimaging Initiative*

*Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at: [http :
//adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf](http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf)

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go here. Keywords: XXX, XXX, XXX

Data collection and sharing for the Alzheimer's Disease Neuroimaging Initiative (ADNI) is funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organization is the Northern California Institute for Research and Education. In the past, ADNI has also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of Health Research, and private sector contributions through the Foundation for the National Institutes of Health (FNIH) including generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC.; Johnson & Johnson Pharmaceutical Research & Development LLC.; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC.; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

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1 Introduction

1.1 Background

According to the World Health Organization (2025), Alzheimer’s disease (AD) is the most common neurodegenerative disorder and a leading cause of dementia around the world with no definitive cure (WHO, 2025). The 2011 National Institute on Aging-Alzheimer’s Association (NIA-AA) diagnostic guidelines describe the disease as a continuum with three stages: a symptom-free stage (pre-clinical AD), a pre-dementia stage (mild cognitive impairment (MCI) due to AD) and a dementia stage (dementia due to AD) (Carrillo et al. 2013, p. 596). Pathophysiological changes due to AD can begin up to 20 years before clinical symptoms appear, highlighting the crucial importance of early diagnosis and timely intervention to potentially slow the progression of the disease (Carrillo et al., p. 595).

In the context of AD the term “biomarker” is commonly used and is defined as “a characteristic that is objectively measured and evaluated as an indicator of normal biological processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention.” (Biomarkers Definitions Working Group, 2001, p. 91).

AD pathophysiology has typically been detected by positron emission topography (PET) to confirm the presence and degree of amyloid plaques in the brain or by quantification of biomarkers implicated in AD pathology in cerebrospinal fluid (CSF) (Mattsson et al., 2017, p. 557). The former method involves radiation and the latter requires a lumbar puncture in order to extract the CSF. WHO published a report in 2024 highlighting the promising progress made in recent years regarding the use of blood-based biomarkers to support AD diagnosis (WHO, p. 6). Specifically, recent breakthroughs in technology (Rissin et al., 2010) have allowed for quantification of biomarkers at much smaller concentrations than previous methods, typically being up to 1000 times more sensitive (Chang et al., 2012, p. 2). Using this method, biomarkers can be detected peripherally from their origin where concentrations are much smaller. These advances mean that more invasive procedures can be substituted with blood tests.

There are several core biomarkers used in the diagnosis and staging of AD (Jack et al., 2024), such as tau protein phosphorylated at the threonine-181 position (p-tau181), with concentrations expected to be higher in individuals with AD pathology (Wong et al., 2012, p. 815). Neurofilament light chain (NfL), a marker of neuronal injury not specific to AD (Jack et al., p. 5158), is also of importance. NfL is usually present inside neurons, and therefore concentrations are expected to be higher in individuals with neuronal injury symptomatic of AD (Clark, C., 2021, p. 7). Research suggests that NfL may be used as a prognostic marker of cognitive decline which supports the diagnosis of a myriad of diseases, including frontotemporal dementia, whilst p-tau181 is more useful for differentiating AD from other neurodegenerative conditions (WHO, 2024, p. 7).

These two blood-based biomarkers, along with various other biomarkers and clinical measures related to AD, are the focus of the *Alzheimer’s Disease Neuroimaging Initiative* (ADNI),

whose main goals are to provide open-access data to researchers worldwide and to improve the diagnosis of AD (ADNI, n.d.-a). Since 2004, researchers across more than 60 clinical sites in the USA and Canada have continuously collected and analyzed data throughout the different phases of the study. Blood samples from the ADNI cohort have been retrospectively analysed using the recently developed quantification method in order to measure p-tau181 and NfL concentrations (Zetterberg & Blennow, 2020; Blennow, 2018).

1.2 Research Context

Recent research has applied a range of prognostic models to estimate disease progression and related adverse outcomes in AD using biomarker data. One study from 2020 used $\log_{10}$ transformed baseline plasma p-tau181 values in a Cox proportional hazards (Cox PH) model and found it to be a strong predictor of conversion across groups, with hazard ratios ranging from approximately 2.5 to 3.8 depending on cognitive status (Janelidze et al., 2020, 388). Adjusting for age, sex and education, baseline plasma p-tau181 levels were strongly associated with increased risk of progression to Alzheimer’s dementia, with hazard ratios around 3-4 per standard deviation increase. A Cox PH model was estimated with additional $\log_{10}$ transformed biomarker levels, including plasma NfL. The effect of p-tau181 remained significant, with only a small reduction in effect size, whereas NfL showed no meaningful contribution to risk prediction with hazard ratios close to 1 (Janelidze et al., pp. 385, 388).

The joint modelling framework extends the Cox PH model by linking a longitudinal model which accounts for the effect of an endogenous time-dependent covariate measured with error, such as a biomarker (Rizopoulos, 2012, p. vii). This approach has been applied to model the relationship between repeated plasma p-tau181 and NfL measurements on the conversion risk to AD (Yuan et al., 2024, p. 1). Higher longitudinal p-tau181 levels were found to be associated with a significantly increased risk of progression, with the joint model estimating a hazard ratio of approximately 5.83 for p-tau181 and 4.18 for NfL (Yuan et al., p. 1). Combining p-tau181 and NfL measurements in a multivariate joint model lead to improved performance (Yuan et al., pp. 1, 3). Several studies have examined the association between plasma p-tau181 and NfL levels and baseline measures such as cognition score or CSF biomarker levels using longitudinal models (Mattsson et al., 2019; Chen et al., 2021).

The simultaneous modelling of multivariate longitudinal biomarkers and clinical endpoints when heterogeneity is highly suspected has been handled by joint latent class mixed models (JLCMMs) (Proust-Lima et al., 2023, p. 3996). JLCMMs model complex disease progression whilst offering a method for the exploration of new pathological hypotheses and identification of subphenotypes. One application of a multi-dimensional JLCMM captured heterogeneity in the trajectories of three cognitive domains (memory, language, and executive functioning) from the time of Alzheimer’s dementia diagnosis (Scollard et al., 2026, p. 383). They identified five latent classes differentiated by overall level of impairment (low-high) and rate of decline (slow-fast). Within each latent class, the pattern of decline was relatively consistent across cognitive domains, suggesting parallel deterioration across functions rather than domain-specific

divergence. In addition, class membership was associated with several key variables, such as APOE genotype, sex, race, education and neuropathological burden (Scollard et al., p. 383). Despite these advances, such modeling approaches have rarely been applied to longitudinal plasma biomarkers in AD research.

Various models have utilised longitudinal p-tau181 and NfL values with the aim of modeling AD progression, however, the question arises as to which of these modeling approaches has greatest predictive performance. Few studies have systematically compared the previously mentioned modeling approaches within a unified framework in AD research, specifically focusing on plasma biomarkers such as NfL and p-tau. Moreover, previous findings suggest that the observed associations with disease progression may vary between these biomarkers depending on how the longitudinal information is modeled.

Additionally, an important consideration is whether to include all participants with repeated biomarker measurements. Although the two previously mentioned studies included all participants regardless of the number of measurements per individual, this is not always the case. For example, a study examining different cognitive biomarkers to predict progression to AD explicitly states: “Participants were included if they had a baseline diagnosis of MCI and at least three follow-up visits with complete longitudinal cognitive assessments” (Guo et al., 2026, p. ?). Similarly, in a study on type 1 diabetes and primary biliary cirrhosis (PBC), the joint model analysis was restricted to participants with recorded height and weight and at least three biomarker measurements (Xu, 2021, p. 9). Another study, focusing on cardiovascular events, restricted their joint model analysis to the cohort of 3784 patients who had at least 10 measurements of the selected biomarker (Wang et al., 2024, pp. 591-592).

In other words, some studies include all available participants, while others impose minimum requirements on the number of repeated measurements, thereby excluding individuals with only a single observation. This variation in inclusion criteria reflects an ongoing methodological discussion regarding how longitudinal data should be handled in practice. For example, in the previously mentioned study, such restrictions were applied to ensure that the approximation method did not lead to “non-ignorable biases” (Wang et al., p. 591). On one hand, participants with only a single observation do not provide information about change over time within individuals. On the other hand, their observations at a single time point still contribute useful information to the analysis by informing estimates of variability and between-subject effects (Fitzmaurice, Laird & Ware, 2011, chapter 8, pp. 214-215).

The choice of inclusion strategy also has important implications for statistical efficiency and validity. Analyses restricted to individuals with complete data are generally less efficient than methods that use all available information, resulting in a loss of precision. This loss depends on the pattern and distribution of missing data across subjects and time points, as well as the extent to which missingness is related to observed outcomes. Importantly, missing data may under certain circumstances also introduce bias, potentially leading to misleading inferences about changes in the mean response. Consequently, the mechanism underlying missing data is a critical consideration in longitudinal analyses, as it determines whether exclusion of incomplete cases may distort results (Fitzmaurice, Laird & Ware, chapter 17, p. 491).

1.3 Research Aims

This study investigates the prediction of progression to AD-related dementia using longitudinal blood biomarker data, comparing traditional and advanced statistical modelling approaches, different biomarkers, and the impact of the number of repeated measurements per individual. More specifically, the research questions are:

1. Which of the considered modelling approaches provides the best prediction of progression to AD-related dementia using longitudinal blood biomarker data?
2. How do these associations differ between p-tau181 and NfL?
3. How does the precision of estimated longitudinal trajectories and their associations with time to AD-related dementia vary with the number of repeated measurements per individual?

Alternativ:

1. Which modelling approach (Cox proportional hazards model with baseline biomarker values, joint models with longitudinal biomarker data, or latent class joint models) provides the best prediction of progression to Alzheimer’s disease-related dementia?
2. Which biomarker strategy (NfL, p-tau181, or a combination of the biomarkers) provides the best prediction of progression to Alzheimer’s disease-related dementia?
3. How does the precision of estimated longitudinal trajectories and their associations with time to AD-related dementia vary with the number of repeated measurements per individual?

2 Methodology

To address the research questions, a series of statistical models will be implemented. In all models, time to progression to dementia due to AD, defined by the first recorded diagnosis of dementia, is the survival outcome.

First, Cox PH models will be used to assess associations between baseline biomarker levels and progression risk. In these models, p-tau181 and NfL will be included either separately or jointly to allow comparison of their individual and combined effects. Second, univariate joint models will be applied to link each biomarker’s longitudinal trajectory to the survival outcome. Third, multivariate joint models will jointly model both biomarker trajectories to capture their combined and potentially correlated effects on disease progression. Finally, JLCMMs will be used to identify subgroups with distinct longitudinal patterns and associated progression risks. Both univariate and multivariate latent class models will be evaluated.

All models will additionally be adjusted for baseline age, sex, and APOE- ϵ 4 status. In the joint models, these covariates will be included in both submodels to account for potential confounding when estimating the association between biomarker trajectories and disease progression.

That being said, since the longitudinal submodels will already include a time variable, namely years since baseline, age at visit will not be included in the LME analyses; its inclusion would be redundant given that it is determined by baseline age and follow-up time. The significance level will be set at 0.05 for all analyses.

Subsequently, the different modelling frameworks will be compared, as well as the estimated associations of p-tau181 and NfL with disease progression. This will be done by evaluating their predictive performance. To ensure an unbiased assessment of model performance and generalisability, the data will be split into in-sample and out-of-sample sets, where the in-sample data are used for model development and the out-of-sample data are reserved for evaluation.

Model performance will be assessed using time-dependent AUC and Brier score for dynamic prediction. For prognostic models predicting progression from MCI to AD dementia, a 2–4 year follow-up is most commonly used (Xia et al., 2025, p. 213). This reflects typical disease progression, where around one-third of individuals with MCI due to AD develop dementia within five years (Alzheimer’s Association, n.d.). Based on this, a landmark time of 1 year and prediction horizons of 2–6 years will be used to capture short- to mid-term progression.

In addition, a precision analysis will be conducted to assess how the number of repeated measurements per individual influences the estimation of longitudinal trajectories and their associations with the survival outcome. This will be evaluated by comparing results between individuals with at least one repeated biomarker measurement and those with at least three repeated measurements, and by assessing differences in model stability and parameter precision.

There is a wide range of biomarkers for AD that could be investigated in this context. However, given the relatively recent focus on plasma-based biomarkers in research, their inclusion was considered timely and relevant. Accordingly, the two biomarkers included in the analyses are p-tau181 and NfL. They were selected due to their established relevance for predicting dementia and AD progression, as well as the availability of rich longitudinal data with repeated measurements and overlapping follow-up information. In addition, they reflect two distinct pathological processes, making them suitable for comparative analyses of disease progression.

Furthermore, additional modelling approaches could have been considered given the aims of the study. However, due to the limited time and scope of this thesis, including more modelling frameworks would have reduced the ability to study each approach in enough detail. Therefore, a smaller set of models was selected to allow for a more detailed and meaningful comparison. The chosen models were based on their methodological relevance and their ability to capture different aspects of the data, providing an overview of different approaches for modelling longitudinal biomarker data and survival outcomes.

The modelling approaches will be applied to data from the ADNI dataset, in accordance with its data use and access policies (ADNI, n.d.-b).

Data used in the preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The current goals include validating biomarkers for clinical trials, improving the generalizability of ADNI data by increasing diversity in the participant cohort, and to provide data concerning the diagnosis and progression of Alzheimer’s disease to the scientific community. For up-to-date information, see adni.loni.usc.edu.

3 Statistical Theory

3.1 Survival Analysis

All of the modeling approaches considered in this study are grounded in the survival analysis framework, where the primary aim is to analyze the time until a prespecified event of interest occurs. Here, time refers to the duration from baseline, defined as the beginning of follow-up, until the occurrence of the event, and the event denotes a specified outcome of interest (Kleinbaum & Klein, 2012, p. 4). In this study, the event is defined as the first dementia diagnosis, and ptau-181 and NfL levels as well as diagnosis status are available at baseline.

Additionally, individuals with AD-dementia are either considered as right-censored, meaning that they do not reach the event during the follow-up period, or as interval-censored, meaning that the exact date of the event is unknown (Kleinbaum & Klein, 2012, pp. 7, 286). This classification is due to diagnoses being made yearly, meaning that the exact date of transition to dementia can only be approximated.

3.2 Cox Proportional Hazards Model

To quantify the association between biomarkers and disease progression, relative hazards models are commonly used. These models compare the hazard of progression between individuals with different covariate profiles, and assume that their hazard rates are proportional. Taking root in the Lehmann alternative for comparing survival functions, which assumes that two survival functions, $S_0(t)$ and $S(t)$, are related such that:

$$S(t) = [S_0(t)]^\psi$$

where ψ is the proportional hazards constant (Moore, 2016, pp. 43-44). This relationship can be re-expressed in terms of their hazard functions to give:

$$h(t) = \psi h_0(t)$$

which is the proportional hazards assumption (Moore, p. 55). Further, covariate information is linked to the proportional hazards constant exponentially, yielding the following model:

$$h(t) = h_0(t) \exp \{ \mathbf{x}^T \beta \}$$

where $\mathbf{x}^T$ is a vector of covariates and β is a vector of model parameters. In this analysis, the covariates of interest in the hazard model are age, gender, APOE- ϵ 4 status and baseline p-tau181 and NfL values.

3.2.1 Model Assumptions

- What do we assume about the censoring process?

A key assumption of this model is the assumed proportionality between hazard functions for different values of the specified covariates. This assumption is crucial for the construction of the partial likelihood, which allows the survival distribution to remain unspecified, as it allows the baseline hazard function to be dropped from the partial likelihood. At the same time, minor violations of this assumption are generally unlikely to have substantial effects on inference and parameter estimation (Moore, p. 94).

3.2.2 Concordance Index

The concordance index (C index) is an evaluation measure for the performance of survival models. All pairs of individuals are selected, and the C index evaluates the proportion of pairs who are concordant, where the individual with a higher cumulative incidence, calculated as $1 - S(t)$, has the shorter time-to-event (Seo Young Park et al., 2017, p. 1701).

3.3 Joint Modelling of Longitudinal and Survival Data

In the joint model of longitudinal and survival data, the hazard function at time t is directly dependent on the value of the biomarker level at time t . In AD research specifically, biomarkers such as p-tau181 and NfL are measured repeatedly over time, and their relationship between their trajectories and disease progression is analysed.

The longitudinal process - modelled by a linear mixed-effects model - and the survival process - modelled by a proportional hazards model - are linked by a specified association structure. This structure can link the survival process to the current value of the biomarker estimate only, or include the current rate of change of the biomarker trajectory. For instance, if both the current value and current rate of change are specified in the proportional hazards model, this becomes:

$$h(t|M_i, \mathbf{v}_i) = h_0(t) \exp \{ \gamma^T \mathbf{w}_i + \alpha_1 m_i(t) + \alpha_2 m'_i(t) \} \quad m'_i(t) = \frac{d}{dt} m_i(t)$$

where α_1 and α_2 link the current value of the biomarker, $m_i(t)$, and the current rate of change, $m'_i(t)$, respectively to the hazard function. Further, γ^T is a vector of log-hazard ratios and $\mathbf{w}_i$ is a vector of time-independent covariates for the i^{th} individual.

This univariate framework is applied to the longitudinal p-tau181 and NfL measurements separately to model their trajectories jointly with survival outcomes. **Multivariate joint models extend the standard joint modeling framework by allowing two or more longitudinal biomarkers to be modeled jointly while being linked to a shared survival outcome describing progression to AD or dementia. Therefore, both p-tau181 and NfL are then considered simultaneously, as they capture complementary aspects of neurodegeneration and Alzheimer-related pathology. *** källa? Ignoring their correlation may lead to biased estimation (Xu, 2021, p. 6)**

3.3.1 Linear Mixed-Effects Submodel

The longitudinal component of the joint model is specified by the linear mixed-effects (LME) model (**LAIRD & WAIRE**), which captures both population-level trends and individual variability in biomarker trajectories.

Longitudinal data consists of repeated measurements from the same individuals, whereby the assumption of independence between observations is illogical. In this analysis, the levels of p-tau181 and NfL for an individual at a specific time point, t , are correlated with the measurements made on prior and subsequent occasions. Therefore, the responses from an individual are expected to be more highly correlated than those between individuals. The modelling of this type of data, where the data can be considered grouped and where there is non-independence between observations, is handled well by LME models.

Following the notation of Rizopoulos (p. 54), the biomarker process for p-tau181 and NfL can be considered such that:

$$y_i(t) = m_i(t) + \epsilon_i(t) \quad \epsilon_i(t) \sim N(0, \sigma^2)$$

$$m_i(t) = \mathbf{X}_i^T \beta + \mathbf{Z}_i^T \mathbf{b}_i \quad \mathbf{b}_i(t) \sim N(0, \sigma^2)$$

where $y_i(t)$ is the observed biomarker measurement at time t for individual i . Assuming that the true biomarker value is not observed due to measurement and other errors, $m_i(t)$ then is the trajectory function which represents the true unobserved value of the biomarker at time t for patient i . The fixed effects are described by $\mathbf{X}_i^T \beta$ and the random effects by $\mathbf{Z}_i^T \mathbf{b}_i$, where β are population-specific parameters and $\mathbf{b}_i$ are subject-specific parameters (Verbeke & Geert

Molenberghs, 2000, p. 23). These subject-specific parameters handle the non-independence of the repeated measurements. Combining the true biomarker trajectory function with the error term gives:

$$y_i(t) = \mathbf{X}_i^T \beta + \mathbf{Z}_i^T \mathbf{b}_i + \epsilon_i(t) \quad \mathbf{b}_i(t) \sim N(0, \sigma^2), \quad \epsilon_i(t) \sim N(0, \sigma^2)$$

The LME model can be generalised when the response variable does not follow an approximately normal distribution. In this case, some link function, $g(x)$, links the expected value given the random effects to the linear predictor, which takes into account both the fixed and random effects (Rizopoulos, pp. 139 - 140). This approach is tested, as p-tau181 and NfL measurements are not normally distributed.

(LAPLACE vs. MCMC in different packages - estimation)

3.3.2 Survival Submodel

Assuming the survival submodel is defined as a Cox PH model, the longitudinal biomarker process is incorporated with the survival process, given a current value only association structure, such that:

$$h(t|M_i, \mathbf{v}_i) = h_0(t) \exp \{ \gamma^T \mathbf{w}_i + \alpha m_i(t) \} \quad M_i = \{m_i, 0 \leq s \leq t\}$$

where M_i represents the true unobserved biomarker profile up to time t .

- Time-to-diagnosis process
- Parametric baseline hazard: **flexible parametrics, splines??**
- OR leave undefined -> Cox PH model

MORE... Weibull in estimation.

3.3.3 Model Assumptions

A core assumption of the joint model for longitudinal and survival data is full conditional independence between the longitudinal outcome and the survival outcome. All dependence is assumed to be explained by the random effects, resulting in independence of the longitudinal and survival outcomes and the repeated biomarker measurements are independent of each other. (**REF**)

3.3.4 Estimation - Frequentist vs Bayesian Approach

The primary estimation approach that has been proposed for joint models is (semiparametric) maximum likelihood (Rizopoulos, 2012, p. 62). However, a key difficulty in joint modelling is that the estimation process can be very computationally demanding and may sometimes be infeasible (Wu et al., 2008).

A Bayesian alternative treats parameters as random variables and combines prior distributions with observed data to obtain posterior distributions, in contrast to the frequentist view where parameters are fixed but unknown (Carlin & Louis, 2008). This approach can simplify inference in complex models, and with weakly informative priors, it often yields results similar to maximum likelihood estimation (Brown & Ibrahim., 2003, p. 222).

Using the R package Jmbayes, joint models can be fitted via Markov chain Monte Carlo (MCMC) algorithms (Rizopoulos, 2018). MCMC generates dependent samples from the posterior distribution using a Markov chain, where each step depends only on the current state. As the chain evolves, it converges to its stationary distribution, allowing the generated samples to be treated as representative of the posterior distribution and used for inference via Monte Carlo integration (Hamra, MacLehose, & Richardson, 2013, p. 628).

In Jmbayes, weakly informative ones are used, allowing stable estimation while letting the data primarily drive inference. The framework also accommodates multivariate settings, by introducing additional association parameters and regularising priors to handle higher-dimensional parameter structures (Rizopoulos, 2018).

3.3.5 The Gelman-Rubin Statistic

Convergence within the framework of Bayesian joint modeling can be evaluated by running multiple chains from different starting values (Peng, 2022, chapter 7.4.2). All chains are expected to eventually reach the stationary distribution and become indistinguishable. Convergence can be assessed by comparing variation between chains to variation within chains, with small between-chain variation indicating that the chains are similar (Peng, chapter 7.4.2).

The Gelman-Rubin statistic R formalizes this idea (Peng, chapter 7.4.2). Consider J Markov chains, each with samples $x_1^{(j)}, x_2^{(j)}, \dots$ from the j -th chain. For each chain, remove the first D samples as “burn-in” and retain the following L samples: $x_D^{(j)}, x_{D+1}^{(j)}, \dots, x_{D+L-1}^{(j)}$. One common choice is to set $D = L$.

Then calculate the chain mean for each chain,

$$\bar{x}_j = \frac{1}{L} \sum_{t=1}^L x_t^{(j)}$$

the overall mean across all chains (the grand mean),

$$\bar{x}_{\cdot} = \frac{1}{J} \sum_{j=1}^J \bar{x}_j$$

the between-chain variance,

$$B = \frac{L}{J-1} \sum_{j=1}^J (\bar{x}_j - \bar{x}_{\cdot})^2$$

the within-chain variance for each chain,

$$s_j^2 = \frac{1}{L-1} \sum_{t=1}^L (x_t^{(j)} - \bar{x}_j)^2$$

and finally, the average within-chain variance,

$$W = \frac{1}{J} \sum_{j=1}^J s_j^2$$

Subsequently, the Gelman-Rubin statistic is:

$$R = \frac{\frac{L-1}{L}W + \frac{1}{L}B}{W}$$

As the number of iterations L increases and the between-chain variation B decreases, R approaches 1. Thus, chains are typically run until R is near 1. Being a unit-free ratio, R provides a convenient summary of convergence without requiring specification of a particular parameter, making it a practical tool for monitoring MCMC chains before any inference is made (Peng, chapter 7.4.2).

3.4 Joint Latent Class Mixed Model

While standard joint models assume a homogeneous population, AD is highly heterogeneous, with patients exhibiting different patterns of cognitive decline and biomarker evolution. JLCMMs address this by assuming that the population consists of a finite number of unobserved subgroups (latent classes), each characterized by distinct longitudinal trajectories and survival risks.

This model links a longitudinal biomarker process and a survival process whilst assuming that the population, N , is heterogeneous and made up of G homogeneous and unobserved latent classes (Kyheng, Babykina & Duhamel, 2025, p. 2). Individuals within latent classes are assumed to

share longitudinal biomarker trajectories and survival risks (Proust-Lima et al., 2018, p. 5) and the classes are not described by the observed covariates (Rizopoulos, 2012, p. 122). The multivariate JLCMM extends the joint latent class modeling framework by allowing multiple longitudinal biomarkers to be modeled simultaneously within each latent class, while linking their joint evolution to a class-specific risk. In this analysis, the univariate JLCMM will be fit on the longitudinal p-tau181 and NfL measurements separately, and then by compared to the multivariate model for both biomarkers, in order to assess which method more accurately models the class-specific hazard of progression to dementia due to AD, and whether, in fact, any latent classes can be distinguished at all.

3.4.1 Class Submodel

The JLCMM consists of three submodels; a multinomial logistic submodel, a linear mixed-effects submodel and a proportional hazards submodel. Multinomial logistic regression models classes, such that:

$$\pi_{i,g} = Pr(c_i = g | \mathbf{X}_i) = \frac{e^{\mathbf{X}_i \beta_g}}{\sum_{l=1}^G e^{\mathbf{X}_i \beta_l}}$$

where $\mathbf{X}_i$ is the set of covariates for the i^{th} individual and β_g is the set of coefficients for the g^{th} group. If there are no prior assumptions about the latent classes, then there will be no covariates in the multinomial logistic regression model (Kyheng, Babykina & Duhamel, p. 2), which further highlights the assumption that the classes are not described by observed covariates (Proust-Lima et al). The class submodel then simplifies to:

$$\pi_{i,g} = Pr(c_i = g) = \frac{e^{\beta_{0,g}}}{\sum_{l=1}^G e^{\beta_{0,l}}}$$

where $\pi_{i,g}$ is the probability that the i^{th} individual belongs to latent class g . Linking the class submodel to the L submodel and the survival submodel, assuming only a random intercept, results in the following:

$$h(t|c_i = g) = h_{0,g}(t) \exp \{ \gamma^T \mathbf{w}_i + \alpha_1 m_i(t) \}$$

3.4.2 Model Assumptions

In JLCMM, the marker and the time-to-event are considered to be conditionally independent given the latent classes and the random effects in the longitudinal model account for the correlation between repeated measures (Proust-Lima et al., 2018, p. 122), whereas in regular joint models the marker and time-to-event are assumed to be conditionally independent given the random effects (Proust-Lima et al., p. 62-63). The conditional independence assumptions for the JLCMM are then as follows;

1. Correlations between observations of the longitudinal biomarker are modelled by the random effects, b_i , in the longitudinal model.
2. The relationship between biomarker trajectory and the time-to-diagnosis is captured completely by the latent class indicator, c_i .

There are various methods for assessing the conditional independence assumption, but simulations have demonstrated the superior performance of the score-test statistic in evaluating residual independence between the longitudinal and survival processes, even against two misspecified alternative hypotheses (Jacqmin-Gadda et al., 2010, p. 17).

The score test assumes the null hypothesis of conditional independence, that is, all correlation between the biomarker trajectory and time-to-event is captured entirely by group membership. Given the class membership is known, there is assumed to be no remaining dependence between the biomarker response and the time-to-event response. Assuming a model with a current value association structure, the association parameter, α_1 , is assumed to be zero under the null hypothesis (Jacqmin-Gadda et al., p. 13), such that:

$$H_0 : \alpha_1 = 0 \quad H_A : \alpha_1 \neq 0$$

Under the null hypothesis, the score test statistic, U , is χ^2 distributed with degrees of freedom equal to the number of shared-random effects in the model (Proust-Lima et al., p. 8). Therefore, a significant test statistic supports the alternative hypothesis that there is residual dependence between the biomarker and time-to-event responses.

3.4.3 Estimation

The log-likelihood for the JLCMM is much more tractable than for the regular joint model (Rizopoulos, p. 123), making estimation less computationally demanding. Log-likelihood estimation for JLCMM involves summing over the latent classes (Proust-Lima et al., 2014, p. 5), which is computationally advantageous when compared with the estimation of joint models which require mathematical approximation through, for example, the previously defined MCMC algorithm, due to integration over the random effects. For JLCMM, the log-likelihood is maximised using a **Marquardt algorithm**. In model estimation it should be considered

that there may be several local maxima, meaning the algorithm should be run several times using different starting values (Proust-Lima et al., p. 5).

... MARQUADT ALGO ... ???

In the *lcmm* package developed by Proust-Lima et al. for estimating JLCMMs, a Weibull baseline hazard model is estimated by default and it is not possible to estimate a Cox PH model, in order to stay in the full-likelihood framework (Proust-Lima et al., p. 5). Therefore, it is necessary to assess whether the hazard is best modeled by the Weibull or by a flexible baseline hazard model.

MORE... Weibull in estimation.

3.4.4 Class Enumeration

Choosing the number of latent classes, also called class enumeration, is one of the challenges of model selection for joint latent class mixture models. Likelihood ratio tests cannot be performed for nested latent class models with differing number of classes, as regularity conditions which ensure a χ^2 distributed test statistic do not hold (Celeux & Soromenho, 1996, p. 196). Therefore, other metrics must be utilised for this purpose.

One aspect of determining the optimal number of classes is minimization of the Bayesian Information Criterion (BIC) (Schwarz, 1978), which is the preferred criterion for mixture models (Proust-Lima et al., p. 7). RESULTS FROM SIMULATION STUDY (MORGAN ET AL.) Ensuring good discrimination between classes is another important component of selecting the number of classes in the final model. Class discrimination can be assessed by analysis of the posterior class membership probability, given by (Proust-Lima et al., p. 7):

$$\hat{\pi}_{i,g} = P(c_i = g | Y_i, (T_i^*, E_i); \hat{\theta}_G) = \frac{\hat{\pi}_{i,g} f(Y_i | c_i = g; \hat{\theta}_G) \lambda_i(T_i | c_i = g; \hat{\theta}_G) S_i(T_i | c_i = g; \hat{\theta}_G)}{\sum_{l=1}^G \hat{\pi}_{i,l} f(Y_i | c_i = l; \hat{\theta}_G) \lambda_i(T_i | c_i = l; \hat{\theta}_G) S_i(T_i | c_i = l; \hat{\theta}_G)}$$

Reporting the proportion of individuals who have a posterior probability over a certain threshold (Proust-Lima et al., p. 7), typically 0.7, 0.8 or 0.9 is a way to assess the ability of the model to adequately discriminate classes from one another. Mean posterior probabilities for each class can also be reported, with $\mu(\hat{\pi}_g)$ values close to 0.5 showing poor class discrimination.

Class sizes are also important, as classes that are made up of less than 5% of the total sample population may

Entropy

One possible measure of the discriminative power of a latent class structure is relative entropy (Ramaswamy et al., 1993 in Morgan et al., 2016, p. 150), which takes values between 0 and 1 and is calculated as follows:

$$EN(G) = 1 - \frac{\sum_{i=1}^N \sum_{g=1}^G \hat{\pi}_{i,g} \log(\hat{\pi}_{i,g})}{N \log(G)}$$

where G is the number of latent classes specified by the model and $\hat{\pi}_{i,g}$ is the posterior probability that the i^{th} individual belongs to class g . Relative entropy values closer to 1 demonstrate good discriminative ability of the model, where, for one class latent models, $EN(1) = 1$.
***CONCEPTUALLY, WHAT DOES THIS MEASURE???**

Integrated Classification Likelihood Criterion

Both fit and discrimination

Validation

The chosen classes should be clinically meaningful.

4 Evaluation of Predictive Performance

Information criteria such as the Akaike Information Criterion (AIC) are often used to evaluate predictive performance in joint models combining longitudinal and survival data. However, when the main focus is the survival outcome and how well a longitudinal biomarker predicts it, performance is typically assessed using two main approaches: calibration, which measures how closely predicted probabilities agree with observed outcomes, and discrimination, which evaluates how well the model distinguishes between individuals who experience the event earlier versus later (Rizopoulos, p. 194). Time-dependent AUC is an example of a discrimination measure, while the Brier score is a calibration-oriented measure that also reflects overall prediction error over time.

4.1 Time-dependent AUC

ROC analysis is commonly used to evaluate the ability of a marker or model to discriminate between binary outcomes, such as diseased and non-diseased individuals (Zou et al., 2012, p. 3). However, the traditional ROC framework assumes that both disease status and marker values are fixed over time (Kamarudin et al., 2017, p. 1).

In survival settings, this assumption is often violated, as event status changes during follow-up and marker values may vary over time. To account for this, time-dependent ROC curves have been developed, incorporating the timing of events and censoring, and providing a more appropriate framework for assessing prognostic accuracy (Kamarudin et al., 2017, p. 1).

When a time-dependent framework is adopted, the disease status is evaluated at each time point, meaning that sensitivity and specificity may change over the course of the study. Let T_i

denote the time of disease onset for individual i , and let X_i represent a marker value (typically measured at baseline) for individual i , where $i = 1, \dots, n$ (Kamarudin et al., p. 2).

QUESTIONS:

1. **When is Z_i used after defining it?**
2. **What is the difference between δ_i and $D_i(t)$? Does δ_i refer to whether the event happens at all, whereas $D_i(t)$ refers to whether the event has occurred by time t ? If they are different, where is δ_i referenced after being defined?**

The observed time is defined as $Z_i = \min(T_i, C_i)$, where C_i is the censoring time. In addition, let δ_i be an indicator variable that takes the value 1 if the event (disease) occurs and 0 otherwise. Furthermore, let $D_i(t)$ denote the disease status at time t , where $D_i(t) = 1$ indicates presence of disease at time t , and $D_i(t) = 0$ otherwise (Kamarudin et al., p. 2).

In what follows, X is referred to as a marker, although it may also represent a risk score derived from a regression model or another predictive model, including previously published scores (Kamarudin et al., 2017, p. 2).

For a given threshold c , the time-dependent sensitivity and specificity are defined as:

$$Se(c, t) = P(X_i > c \mid D_i(t) = 1)$$

$$Sp(c, t) = P(X_i \leq c \mid D_i(t) = 0)$$

Based on the definitions above, the corresponding receiver operating characteristic (ROC) curve at a given time t , denoted by $ROC(t)$, can be constructed by plotting the time-dependent sensitivity $Se(c, t)$ against $1 - Sp(c, t)$ across all possible thresholds c (Kamarudin et al., p. 2).

The time-dependent area under the curve (AUC) is then defined as:

$$AUC(t) = \int_{-\infty}^{\infty} Se(c, t) d[1 - Sp(c, t)]$$

An AUC value of 1 means perfect ability to distinguish between patients with and without the condition, while an AUC of 0.5 means no discrimination at all, meaning the marker performs no better than random guessing such as flipping a coin (Rizopoulos, p. 196)

4.2 Brier score

The Brier score can be thought of as the mean squared error of a prediction forecast. However, for survival data, the true event status cannot be known for censored individuals. Therefore, an inverse-probability-of-censoring weighting (IPCW) adjusted Brier score has been formulated to account for bias due to censoring, where those with earlier event times are over-represented, leading to underestimation of survival probabilities (Prince et al., 2025, p. 4). IPCW weights are calculated for each individual $i = 1, \dots, n_{sample}$ at each time point t , such that:

$$\omega_i(t) = \frac{\mathbb{I}\{T_i^* \leq t, \delta_i = 1\}}{G_i(T_i^*-)} + \frac{\mathbb{I}\{T_i^* > t\}}{G_i(t)}$$

where $G_i(t)$ is the censoring survival function, which quantifies the probability of surviving a censoring event for the i^{th} individual, T_i^*- is a time point infinitesimally smaller than the observed time T_i^* , and $\mathbb{I}\{\cdot\}$ is an indicator function (Prince et al., pp. 4 - 5). The commonly formulated Brier score:

$$BS = \frac{1}{n_{sample}} \sum_{i=1}^{n_{sample}} (Y_i(t) - \widehat{S}_i(t))^2$$

where $\widehat{S}_i(t)$ is the predicted probability of an event for the i^{th} individual at time t and $Y_i(t)$ is their observed outcome at that time (Seo Young Park et al., 1704), becomes then the IPCW-adjusted Brier score:

$$BS = \frac{1}{\tilde{n}(t)} \sum_{i=1}^{n_{sample}} \left(\frac{\widehat{S}_i(t)^2 \mathbb{I}\{T_i^* \leq t, \delta_i = 1\}}{G_i(T_i^*-)} + \frac{(1 - \widehat{S}_i(t))^2 \mathbb{I}\{T_i^* > t\}}{G_i(t)} \right)$$

where $\tilde{n}(t)$ is the sum of the IPC weight (Prince et al., p. 6). The Brier score takes values from 0 to 1, where a score closer to 0 is given to a model with better predictive ability.

5 Data

The data used in this analysis comes from the ADNI cohort under the ADNI1 and ADNI2 protocols. Measurements of the concentration of p-tau181 and NfL in blood plasma of $n = xxx$ individuals were analysed. In accordance with the ADNI leadership guidelines, which state that there were serious data quality concerns for individuals with a PTID in the format "381_S_10####", checks were run to assure that no observations with these PTIDs were in the data set (ADNI, 2026).

The variables of interest were all retrieved from the ADNIMERGE2 package, found in the ADNI database, which include various datasets all containing different variables related to the study of Alzheimer’s (ADNI, n.d.-b). The inclusion criteria for this study require participants to have recorded diagnostic visits and biomarker measurements of both p-tau181 and NfL. Participants who do not meet these criteria are excluded.

5.1 Variables

5.1.1 Diagnosis status

The variable of primary interest is an ordinal categorical variable representing the diagnostic status of the participants. More precisely, it includes three diagnostic states: cognitively normal (CN), mild cognitive impairment (MCI) and dementia. Each participant has multiple examination dates, meaning that diagnostic status may change over time. Since the study focuses on time to incident dementia, participants are classified as either non-demented or demented for the survival analyses, without distinguishing between CN and MCI states. The variable was retrieved from the DXSUM dataset in ADNIMERGE2.

5.1.2 P-tau181

P-tau181 is a continuous variable representing one of the two plasma biomarkers used to potentially improve prediction of progression to dementia, since, as previously mentioned, it can distinguish Alzheimer’s disease from other neurodegenerative conditions. The variable was retrieved from the UGOPTAU181 dataset in ADNIMERGE2, which has been processed by the University of Gothenburg (ADNI, n.d.-b). However, the unit of measurement is not specified in the dataset documentation.

5.1.3 NfL

NfL is a continuous variable representing the second plasma biomarker included in the analyses, with potential to improve prediction of progression to dementia, as it may, as previously mentioned, serve as a prognostic marker of cognitive decline. It was retrieved from the BLENNOWPLASMANFLLONG dataset in ADNIMERGE2, and the data were processed by the Blennow Lab (ADNI, n.d.-b). As for p-tau181, the unit of measurement is not specified in the dataset documentation.

5.1.4 Age

Additionally, age is included as a covariate when predicting progression to dementia, as risk of dementia is known to increase with age (Alzheimer’s Society, n.d.). The variable was retrieved from the PTDEMOG dataset and was initially recorded as year of birth for each participant (ADNI, n.d.-b). It is then transformed into age at visit and age at baseline, which are more suitable for the analyses.

5.1.5 Gender

Another covariate included when predicting progression to dementia is participants’ sex at birth, as recorded on their original birth certificate, due to reported differences in disease risk and progression between sexes. More precisely, women aged over 80 have a slightly higher chance of developing Alzheimer’s disease compared to men of the same age (Alzheimer’s Society, n.d.). This binary variable includes two categories: male and female (ADNI, n.d.-b). As for age, the variable was retrieved from the PTDEMOG dataset.

5.1.6 APOE4-status

Finally, apolipoprotein- $\epsilon 4$ (APOE- $\epsilon 4$) status is included as a covariate when predicting progression to dementia, due to its well-established role as a genetic risk factor for AD. The $\epsilon 4$ allele of apolipoprotein E (APOE) increases the risk of AD, whereas the - $\epsilon 2$ allele has been shown to have a protective effect (Wong et al.). The variable was retrieved from the APOERES dataset (ADNI, n.d.-b).

Initially coded numerically based on the number of $\epsilon 4$ alleles (ADNI, n.d.-b), the variable is recoded into three ordinal categories: 0 alleles (non-carriers), 1 allele (heterozygous carriers; 2/4 or 3/4), and 2 alleles (homozygous carriers; 4/4).

6 Data Pre-processing and Descriptive Analysis

ADNI documentation: <https://adni.loni.usc.edu/quick-start-guide-asset/anatomy2.html>

6.1 Missing Values

All missing values were removed prior to the analysis to ensure a complete-case dataset and avoid bias introduced by incomplete observations.

6.2 Number of Participants

The inclusion criteria resulted in a sample size of 1191 individuals with at least one biomarker measurement. Additionally, when restricting the sample to participants with three or more biomarker measurements, the sample size was reduced to 704.

6.3 Distributions, Transformation and Baseline Definition

Biomarker examination dates ranged from 1 February 2007 to 6 June 2016, while diagnostic assessment dates spanned 29 September 2005 to 7 November 2025. This means that diagnoses were available several years prior to the first biomarker measurements. Baseline was therefore defined as the first biomarker examination date, ensuring alignment of follow-up time with biomarker onset and comparable longitudinal trajectories across participants. Baseline age ranged from 55 to 94 years, with a median age of 74 years.

Among the 704 participants with at least one biomarker measurement, diagnostic visits per individual ranged from 1 to 20, with a median of 7. In total, there were 3198 cognitively normal (CN), 3621 mild cognitive impairment (MCI), and 505 dementia diagnoses, of which 2206 CN, 2780 MCI, and 491 dementia diagnoses occurred after baseline. Restricting to participants with at least three biomarker measurements, the corresponding numbers were 2469 CN, 3163 MCI, and 463 dementia diagnoses, with 1723 CN, 2489 MCI, and 459 dementia diagnoses post-baseline. Diagnostic visits per individual ranged from 3 to 20, with a median of 8.

The number of biomarker measurements per participant ranged from one to five. Specifically, 66 participants had one measurement, 164 had two measurements, 223 had three measurements, 303 had four measurements, and 178 had five measurements.

Among participants with at least three biomarker measurements, p-tau181 values ranged from 0.4 to 238.3, with a median of 15.5. Similarly, NfL values ranged from 6.9 to 408.0, with a median of 36.5. These descriptive statistics indicate pronounced right-skewness for both biomarkers, motivating the use of transformation to better satisfy the assumption of normality. A similar right-skewed distribution was observed when including all participants with at least one biomarker measurement.

Moreover, when examining observations with large Pearson residuals, specifically those exceeding three standard deviations from the mean, clear individual-level patterns were observed. For example, p-tau181 for one individual increased from 124 to 238 over time, and another individual showed consistently high values, ranging from 110 to 139. Similar patterns were observed for values of NfL. Accordingly, and given that the values fall within plausible biological ranges, these observations were not treated as outliers but were retained in the analysis.

Following log-transformation, p-tau181 exhibited a distribution that more closely approximated normality, as shown in Appendix X. The values became more concentrated around the central region, although a slight residual left-skewness remained. However, this deviation was

not considered substantial. Similarly, Appendix X demonstrates that the log-transformation of NfL resulted in an even closer approximation to a normal distribution.

Additionally, among the participants with at least three biomarker measurements, 315 were female and 389 were male. Regarding genotype status, 83 participants carried no APOE- ϵ 4 alleles, 578 carried one allele, and 43 carried two alleles. When looking at the participants with at least one biomarker measurement, 438 were female and 496 were male. Furthermore, 109 of these participants carried no APOE- ϵ 4 alleles, 772 carried one allele, and 53 carried two alleles.

These distributions support the use of the variables in the subsequent statistical models.

6.4 Processing of Examination Date

Regarding biomarker measurements, 52 participants had two measurements recorded on the same examination date, leading to minor discrepancies depending on whether the first, last, or mean value was used. To ensure consistency, the mean value was selected as the primary approach. Sensitivity analyses showed no substantial differences in model precision.

Furthermore, discrepancies were observed between p-tau181 and NfL examination dates within the same visit. Five cases exceeded 30 days, with differences of up to 231 days, indicating that these did not represent the same examination occasion. To address this, a common examination date was constructed using the midpoint between the two biomarker dates. Diagnostic records occurring more than 50 days prior to the first biomarker measurement were excluded to maintain temporal alignment between clinical and biomarker data.

Subsequently, sensitivity analyses were performed using alternative definitions of baseline, defined as either the earliest biomarker examination date or the first diagnostic examination date. No meaningful differences in model precision were observed, indicating robustness to the choice of time origin.

6.5 Division Into In-Sample And Out-of-Sample Datasets

Additionally, the analyzed data were split so that the in-dataset comprised three quarters of the total data and the out-dataset comprised one quarter.

When restricting the analysis to participants with at least one biomarker measurement, the in-dataset consisted of 701 individuals, among whom 145 experienced the event of dementia, corresponding to 20.69%, while 556 individuals were censored. The out-dataset included 233 individuals, among whom 53 experienced the event of dementia, corresponding to 22.75%, while 180 individuals were censored.

In comparison, when restricting the participants to those with three or more biomarker measurements, the sample size of the in-dataset was 528, among which 128 experienced the event

of dementia, corresponding to 24.24%, while 400 individuals were censored. In the out-dataset, there were 176 individuals, among whom 49 experienced the event of dementia, corresponding to 27.84%, while 127 individuals were censored.

Related to this, participants who experienced dementia right at baseline were excluded from the survival analyses, as they did not contribute valid follow-up time.

6.6 Correlation Between Numerical Model Variables

Finally, a correlation matrix was computed for the numerical variables in the in-sample dataset to assess potential multicollinearity. The results are presented in Appendices X and X. The correlations were generally low to moderate, indicating no issues with multicollinearity and supporting their inclusion in the models.

It is also worth noting that the correlation between log-transformed p-tau181 and NfL was 0.34, indicating a weak positive association. This suggests that the biomarkers share some common variation while largely capturing distinct biological processes. Overall, this supports the use of a multivariate joint modelling framework, as it allows their dependence to be accounted for while evaluating both their individual and combined associations with disease progression.

7 Results

7.1 Cox PH Models

To evaluate the association between biomarkers and progression to dementia due to AD, while adjusting for relevant covariates, Cox PH models were fitted including either baseline p-tau181, baseline NfL, or both biomarkers measured at baseline, restricted to participants with three or more biomarker measurements. Additional covariates were gender, age at baseline and APOE genotype. Initially fitting the model with all levels of genotype - which denotes the specific APOE allele combination - resulted in non-significant coefficients for all levels. Therefore, genotype was collapsed into three levels denoting the presence of the APOE- $\epsilon 4$ allele, taking the following values: 0 alleles (non-carriers), 1 allele (heterozygous carriers; $\epsilon 2/\epsilon 4$ or $\epsilon 3/\epsilon 4$), and 2 alleles (homozygous carriers; $\epsilon 4/\epsilon 4$).

Prior to interpreting the estimated hazard ratios, the proportional hazards assumption was assessed for all covariates to verify that the effects remained constant over time. As the subsequent joint models also require a standard Cox PH model, this assumption was additionally evaluated for this model. The resulting p-values are presented in Appendix X.

At a 5% significance level, most covariates satisfied the proportional hazards assumption across all models, as indicated by non-significant p-values for sex, APOE- $\epsilon 4$ status, and the biomarker

variables. However, age at baseline consistently showed significant p-values, indicating a violation of the assumption. The same pattern was observed for the GLOBAL test, which jointly evaluates the proportional hazards assumption across all covariates and is sensitive to covariate-specific departures from the assumption; it also yielded significant or borderline significant p-values.

Given this violation, age at baseline was excluded from the final Cox PH models. The refitted models therefore yielded the p-values presented in Table 1.

Table 1: P-values for the proportional hazards assumption tests after exclusion of age at baseline in the Cox PH models estimated on participants with at least three biomarker measurements

variable	Standard	p-tau181 baseline	NfL baseline	Both baseline
PTGENDER	0.864	0.875	0.849	0.907
APOE4_status	0.178	0.186	0.134	0.158
GLOBAL	0.324	0.448	0.163	0.266
baseline_value_ptau	NA	0.692	NA	0.623
baseline_value_nfl	NA	NA	0.134	0.108

After excluding age at baseline, the GLOBAL test yielded p-values above the 5% significance level and therefore no longer indicates any violation of the proportional hazards assumption. The other covariates also remained non-significant, suggesting that they jointly do not exhibit systematic time-dependent effects in any of the refitted models.

As the proportional hazards assumption was met in the refitted models, the results can be interpreted in terms of hazard ratios, which are, together with 95% confidence intervals and p-values, presented in Table 2 for the Cox PH models including either baseline p-tau181, baseline NfL, or both biomarkers measured at baseline.

Table 2 indicates that sex was not significantly associated with progression to dementia due to AD in any of the models, with all p-values exceeding 0.05. Despite non-significant coefficients for sex and non-significant likelihood ratio tests (**Appendix X**) these variables were included due to clinical relevance.

In contrast, APOE- ϵ 4 status showed a strong and consistent association with progression risk, with increasing risk observed according to the number of ϵ 4 alleles carried. Individuals carrying one ϵ 4 allele had approximately a twofold increased hazard across models, whereas individuals carrying two ϵ 4 alleles showed a substantially higher risk, corresponding to an approximately three- to fourfold increased hazard. These associations were highly statistically significant across all models. The confidence intervals for individuals with one ϵ 4 allele were relatively narrow, indicating stable estimates, while the wider intervals among individuals with two ϵ 4 alleles suggest greater uncertainty, likely reflecting the smaller number of participants in this subgroup.

Both baseline p-tau181 and baseline NfL were significantly associated with an increased risk of progression in separate models, with p-values below 0.001. However, NfL showed a slightly stronger association with progression risk than p-tau181 (HR = 2.39 vs. HR = 2.04), indicating a modestly higher effect estimate for NfL. At the same time, the confidence interval for NfL was somewhat wider, suggesting slightly greater uncertainty in its effect estimate compared with p-tau181.

In the Cox PH model including both biomarkers simultaneously, the results were similar, with both p-tau181 and NfL remaining significantly associated with progression. The effect estimates were however slightly attenuated compared with the separate models, with p-tau181 showing a hazard ratio of 1.81 and NfL showing a hazard ratio of 1.98. This indicates that both biomarkers retained independent prognostic value. The confidence intervals for both variables were somewhat wider than in the univariable models, reflecting increased uncertainty in the joint model due to shared variance between the biomarkers.

An initial comparison of model performance by assessment of the concordance indices for all cox models was performed. The results presented in Table 3 show that the concordance index evaluates all three models similarly.

Table 2: Concordance indices for the fitted Cox PH models with baseline biomarker values estimated on participants with at least three biomarker measurements

Model	C-Index
Cox ptau-181	0.679
Cox NfL	0.673
Cox ptau-181 & NfL	0.681

Furthermore, when comparing individuals with three or more measurements of p-tau181 and NfL to those with at least one measurement, the results remained robust, with inclusion of participants with fewer repeated observations increasing statistical power but not changing the direction or interpretation of the estimated effects for the main biological predictors.

More precisely, age at baseline continued to violate the proportional hazards assumption and was therefore excluded from the final models, after which all models satisfied the assumption. Table 4 further demonstrates that hazard ratios, confidence intervals, and patterns of statistical significance remained broadly consistent across models, supporting unchanged conclusions.

Across both model specifications, APOE- ϵ 4 remained the strongest and most stable genetic predictor of progression. Hazard ratios were very similar, with a slight increase in the larger cohort, from about 2.0 to 2.2 for one allele and from about 3.6 to just above 4.0 for two alleles. The confidence intervals were of similar width across both analyses, suggesting stable uncertainty in the estimates and consistent precision.

For the biomarkers, both ptau and NfL were consistently associated with progression, but NfL showed stronger effects and slightly lower uncertainty than ptau across both cohorts. For

example, in single-biomarker models, NfL showed higher hazard ratios than ptau and generally slightly narrower confidence intervals, indicating more precise estimation of its association.

When both biomarkers were included simultaneously, the same pattern remained, with NfL consistently showing higher hazard ratios than ptau in both model specifications. Effect sizes were slightly lower compared to single-biomarker models, and confidence intervals were somewhat wider, consistent with shared variance between biomarkers, but the relative ranking between ptau and NfL was unchanged.

In contrast, sex showed no association with progression in any of the models, with p-values above 0.05.

When defining the event time as the midpoint between the diagnosis exams when a positive dementia diagnosis was given, the results of the proportional hazard test gave the same conclusions (**appendix: table??**) as well as mostly very similar hazard ratios for the cox PH and joint models (**appendix: table?? & table??**). However, the effect of the log-transformed p-tau181 and NfL measurements were somewhat reduced in the univariate joint models when the midpoint was used as the event time (**table?? vs. appendix: table??**). $\widehat{R}$ values for both the survival submodel and the longitudinal submodel were similar to the models estimated with the original exam date as the event time (**tables ?? & ?? vs. appendix tables ?? & ??**)

7.2 Joint Models

The evaluation of the association between biomarkers and progression to dementia due to AD, while adjusting for relevant covariates, was further extended by fitting joint models that incorporated the longitudinal trajectories of p-tau181 and NfL with the time-to-diagnosis data. For the longitudinal component, an LME model was fitted separately for p-tau181 and NfL. As for the Cox models, these models were first estimated in participants with at least three biomarker measurements.

The choice of random effects structure was limited due to the *JMBayes2* package currently only supporting those including a random slope. Therefore, all LME models were fit with a random intercept and slope. However, models with a simple structure with only a random intercept and an extended structure including a random slope were fit using maximum likelihood estimation in order to perform likelihood ratio tests. The results show that the extended structure was a better fit for NfL, but a simpler structure would have been preferable for p-tau181 (**appendix X**).

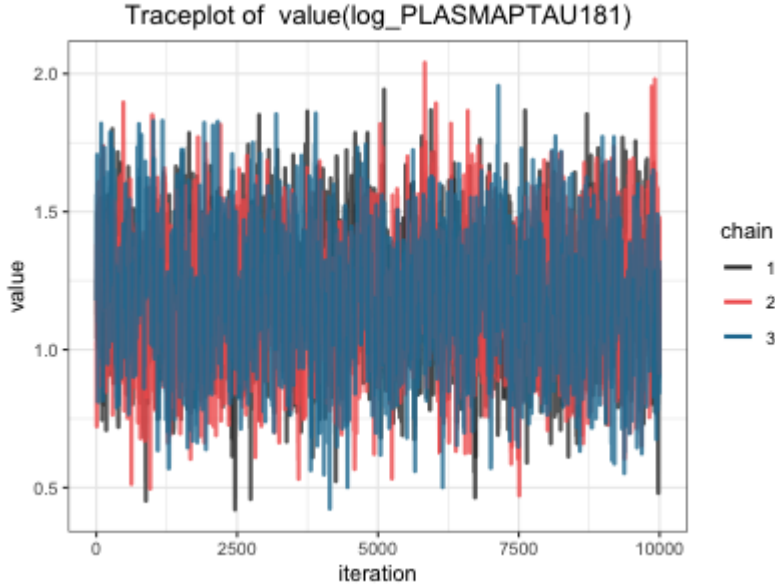
Plotting the residuals against the fitted values for the LME model of p-tau181 showed no clear evidence of heteroskedasticity, as shown in Appendix X. The slight deviations in the tails of the Q-Q plot, shown in Appendix X, were also not considered to be of concern. A similar pattern was observed for the LME model of NfL, where the plot of residuals against fitted values in Appendix X showed even less evidence of heteroskedasticity compared to the corresponding

plot for p-tau181. The Q-Q plot for NfL in Appendix X showed similar results to those observed for p-tau181. Generalised linear mixed effects models with different link functions were also fit and their residuals analysed, however none of these models demonstrated a better fit than the log-transformed model. Consequently, these were considered to provide an adequate fit to the data, suggesting that they are suitable for use within the joint modeling framework.

Thus, joint models including longitudinal measurements of p-tau181, NfL, or both biomarkers were fitted. For each model, three MCMC chains were used, with 11,000 iterations and 1,000 burn-in iterations for the simple joint models, and 20,000 iterations with 2,000 burn-in iterations for the multivariate joint model.

To ensure proper model convergence, trace plots of the α -parameter were examined for each joint model, starting with those for the joint model of p-tau181, presented in Figure 1.

Figure 1: trace plot of the α -parameter for the joint model of p-tau181 estimated on participants with at least three biomarker measurements.



The trace plot shows that the three MCMC chains fluctuate around a stable mean value throughout the iterations, without any clear trend or drift. This suggests that the chains have reached a stable region. In addition, the chains overlap well and mix with each other, indicating that they are sampling from the same posterior distribution. Although some random variation is present, this is expected in MCMC sampling. Overall, the plot supports good convergence.

A similar pattern was observed for the joint model of NfL, as shown in Appendix X. The chains likewise fluctuated around a stable mean value with good mixing, indicating convergence of the α -parameter comparable to the joint model of p-tau181.

Finally, the multivariate joint model also exhibited a similar overall behaviour, as shown in Appendix X. Compared to the single-biomarker models, however, the trace plot for the multivariate joint model shows a slightly wider range of α -values, indicating a somewhat larger variability in the α -parameter. This suggests increased uncertainty in the parameter estimates, which is expected given the higher complexity of the model. Despite this, the chains still fluctuate around a stable mean value with no visible trend or drift, and they continue to overlap and mix well. Thus, all three models appear to have converged adequately.

Additionally, convergence of the models was assessed by examining the $\widehat{R}$ values of the survival outcome and the longitudinal outcome in each joint model, with the survival $\widehat{R}$ values presented in Table 3.

Table 3: $\widehat{\mathbf{R}}$ values of the survival outcome in the joint models

Variable	PTau	NfL	Multi
PTGENDERMale	1.000	1.000	1.000
APOE4_status1 allele	1.004	1.005	1.002
APOE4_status2 alleles	1.000	1.002	1.001
value(log_PLASMAPTAU181)	1.003	NA	1.002
value(log_PLASMA_NFL)	NA	1.003	1.002

The $\widehat{R}$ values for the survival outcomes are all very close to 1, ranging from 1.000 to 1.007 across all joint models. This indicates good convergence of the MCMC chains for the survival components and confirms the convergence patterns observed in the α -parameter trace plots, with no evidence of convergence issues.

While Table 3 presents the $\widehat{R}$ values for the survival outcomes, the corresponding $\widehat{R}$ values for the longitudinal outcomes in each joint model are shown in Table 4.

Table 4: $\widehat{\mathbf{R}}$ values of the longitudinal outcome in the joint models

Variable	Long_PTau	Long_NfL	Long_Multi
(Intercept)	1.001	1.000	1.001
time_years	1.024	1.001	1.023
PTGENDERMale	1.000	1.000	1.001
APOE4_status1 allele	1.002	1.001	1.000
APOE4_status2 alleles	1.000	1.000	1.001
sigma	1.006	1.021	1.024

In a similar manner, the $\widehat{R}$ values for the longitudinal outcomes are generally close to 1 across all joint models, indicating good convergence for most parameters. The intercepts and covariate effects show values very close to 1, suggesting stable convergence. However, a higher value is

observed for the time effect in the multivariate model, where $\widehat{R}$ reaches 1.089. This suggests somewhat slower convergence for this parameter in the multivariate model, likely reflecting the increased complexity of the longitudinal structure. Overall, when considered together with the trace plots and survival $\widehat{R}$ values, the results still support adequate convergence of the models.

Subsequently, the hazard ratios from the joint models were estimated and are presented in Table 5, along with 95% confidence intervals and p-values.

Table 5 presents the hazard ratios estimated from the joint models. Similar to the Cox PH models, sex was not significantly associated with progression to dementia due to AD in any of the models, with all p-values exceeding 0.05. In contrast, APOE- ϵ 4 status showed a clear and consistent association with progression risk, with higher hazards observed as the number of ϵ 4 alleles increased. Individuals carrying one ϵ 4 allele had an approximately twofold increased hazard across models, while individuals carrying two ϵ 4 alleles showed a substantially greater risk, corresponding to an approximately three- to fivefold increased hazard. These associations were statistically significant throughout the analyses. The confidence intervals for individuals with one ϵ 4 allele were relatively narrow, indicating stable estimates, while the wider intervals among individuals carrying two alleles suggest greater uncertainty, likely reflecting the smaller number of participants in this subgroup.

Both longitudinal p-tau181 and longitudinal NfL were significantly associated with an increased risk of progression in the separate joint models, with p-values below 0.001. Similar to the Cox PH models, NfL showed a slightly stronger association with progression risk than p-tau181, with hazard ratios of 3.45 and 3.33, respectively. The confidence intervals for both biomarkers were relatively narrow, suggesting reasonably precise and stable effect estimates.

In the multivariate joint model including both biomarkers simultaneously, both p-tau181 and NfL remained significantly associated with progression. However, the estimated effects were attenuated compared with the separate models, with p-tau181 showing a hazard ratio of 2.30 and NfL showing a hazard ratio of 2.36. This attenuation suggests that the biomarkers capture partly overlapping information related to disease progression, while still retaining independent prognostic value. The confidence intervals for both biomarkers were somewhat wider in the multivariate model, reflecting increased uncertainty due to the correlation between the longitudinal biomarker processes.

To assess the robustness of these findings, the same models were subsequently re-estimated in the full cohort including participants with at least one biomarker measurement. Overall, the results remained highly consistent with those obtained in the primary analysis. Residual patterns and Q-Q plots remained broadly unchanged, and model convergence was stable across specifications. As shown in Table X, hazard ratios, confidence intervals, and patterns of statistical significance were also largely consistent with the main results.

In the joint models, APOE- ϵ 4 remained a strong and stable predictor of progression. Hazard ratios were slightly higher in the model including all individuals, increasing from about 1.9 to 2.2 for one allele and from about 3.2 to 3.5 for two alleles, with a further increase to around

4.3 for two alleles in the full model. Confidence intervals remained well separated from one and show substantial overlap between model specifications, supporting the robustness of the genetic association within a joint modeling framework.

For the longitudinal biomarkers, a clearer separation between ptau and NfL was observed. In both model settings, NfL consistently showed a stronger association with progression than ptau. In the first joint model, ptau had a hazard ratio around 3.3, while NfL was around 3.5, and in the more inclusive model these increase to approximately 4.5 for ptau and 5.5 for NfL. NfL also tended to show slightly narrower confidence intervals than ptau, indicating lower uncertainty in its estimated association.

When both biomarkers were included simultaneously, the same pattern remained, with NfL consistently showing higher hazard ratios than ptau in both joint model specifications. Effect sizes were slightly lower compared to models with single biomarkers, and confidence intervals were correspondingly wider, consistent with shared variance between biomarkers.

Sex also remained non-significant across all joint models, with no meaningful differences between model specifications.

7.3 Joint Latent Class Mixed Models

For all models, a Weibull baseline hazard model was chosen and checked against whether a flexible baseline hazard with the same number of parameters resulted in a lower BIC, which was the case for all models. The results were verified using the *gridsearch()* function which allows for better exploration of the parameter space to ensure finding a global instead of a local maximum solution. This check confirmed that using random starts from the one class model was sufficient for finding the global maximum.

Firstly, latent class mixed models were fit on the longitudinal p-tau181 and NfL measurements in order to assess whether latent classes could be distinguished from the biomarker trajectories. None of these models could adequately separate the sample population into meaningful latent classes, suggesting that p-tau181 and NfL biomarker trajectories alone do not hold enough information to be able to distinguish heterogeneous disease sub-types. Multivariate latent class models were estimated on p-tau181 and NfL measurements simultaneously. Again, this model could not discriminate valid latent classes. Interestingly, incorporating survival data into the models resulted in JLCMMs that could meaningfully classify individuals into latent classes, the results of which follow below.

JLCMM of longitudinal p-tau181 measurements and time-to-dementia data

The random effects structure was decided by a likelihood ratio test between a one class model with random intercept and one with both a random intercept and slope, as well as analysis of the BIC. Despite a non-significant test statistic of the likelihood ratio test ($\chi^2 = 1.38, p = 0.50$), a random effects structure with only intercept was chosen due to a lower BIC for the simpler model ($BIC = 4077$) compared with the complex model ($BIC = 4088$). Next, JLCMMs were

fit with up to four classes and analysis of their log-likelihoods, BIC and entropy follow. All parameters in all of the candidate models were significant ($p < 0.001$) and the results are presented in the **appendix (X, X, X, X)**.

The results show that both log-likelihood and entropy increase with additional latent classes, yet the BIC is minimised for the model with two latent classes (**table ??**). Given the inconclusive results from this comparison, the posterior probabilities were additionally analysed in order to decide the number of latent classes. The posterior classification table shows the number of individuals assigned to each class, whereas the mean posterior probabilities show, for individuals in each class, the average probability of being assigned to each class. For example, values for *class1*, *prob1* close to 1 and *class2*, *prob1* close to 0 show good class discrimination.

Table 5: Table 6: log-likelihood, BIC and entropy for JLCMMs with longitudinal p-tau181 measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)	Class4 (%)
Model1	1	-2019.695	4077.005	1.000	100.00	NA	NA	NA
Model2	2	-1989.499	4041.688	0.648	24.43	75.57	NA	NA
Model3	3	-1984.353	4056.474	0.661	71.21	3.79	25.00	NA
Model4	4	-1974.834	4062.511	0.714	67.42	6.63	22.92	3.03

For the two class model (**table ??**), the class sizes are reasonable and the mean posterior probabilities show good class discrimination, meaning the model has high certainty in assigning individuals to their respective classes. Comparison with the three class model (**table ??**) shows somewhat poorer class discrimination as well as a questionably small *class3*, with only 3.79% of individuals being assigned to this class.

Table 6: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal p-tau181 measurements

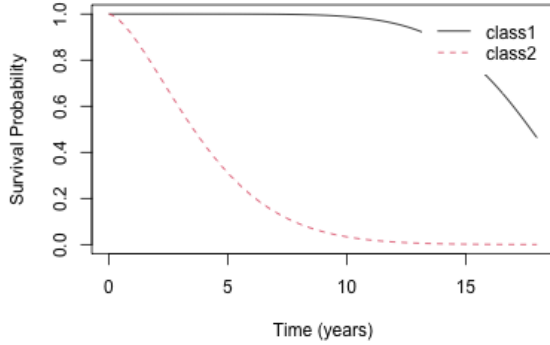
	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9671	0.0329	93.80	91.47	91.47
Class2	0.1204	0.8796	90.98	76.19	52.63

Table 7: Posterior classification table and mean posterior probabilities for the three class JLCMM for longitudinal p-tau181 measurements

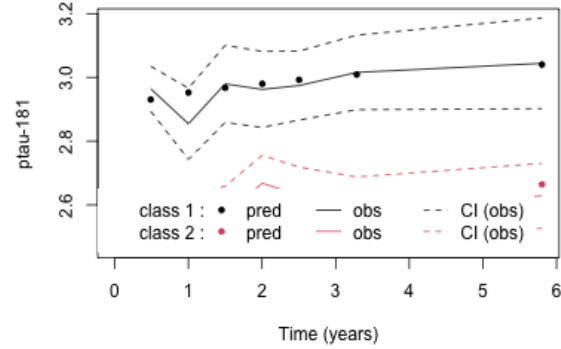
	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8242	0.0521	0.1237	86.17	62.77	28.72
Class2	0.2388	0.7429	0.0183	65.00	45.00	15.00
Class3	0.0433	0.0072	0.9494	93.18	87.88	84.09

Additionally, the score test statistic for the conditional independence assumption was signif-

icant for the one class model ($50.336, p < 0.001$) but upon fitting two latent classes became non-significant ($0.122, p = 0.727$), further evidence that the data supports two latent classes in the sample population. Therefore, the two class model is chosen as the best model for most accurately discerning latent classes from the longitudinal p-tau181 measurements and the time-to-dementia data.



(a) **Figure 4: class-specific survival curves for the two class JLCMM for longitudinal p-tau181 measurements**



(a) **Figure 5: class-specific p-tau181 trajectories for the two-class JLCMM**

Figure ?? shows the class-specific survival curves. The probability of remaining dementia-free reduces sharply for *class2* up until about 5 years, where the curve starts to level out. For individuals in *class1*, the probability of dementia diagnosis remains close to 0 for about 10 years, where the probability of diagnosis increases slowly at first and then, from about 15 years, increases sharply.

JLCMM of longitudinal NfL measurements and time-to-dementia data

Again, the results of a likelihood ratio test between two models was used to determine the random effects structure. This resulted in a significant test statistic ($\chi^2 = 10.5, p = 0.005$), evidence that the simpler model with only a random intercept was a better fit for the data. Three JLCMMs were fit and all model parameters in all three models were significant ($p < 0.001$, **appendix X, X, X, X**).

Table 8: Log-likelihood, BIC and entropy for JLCMMs with longitudinal NfL measurements

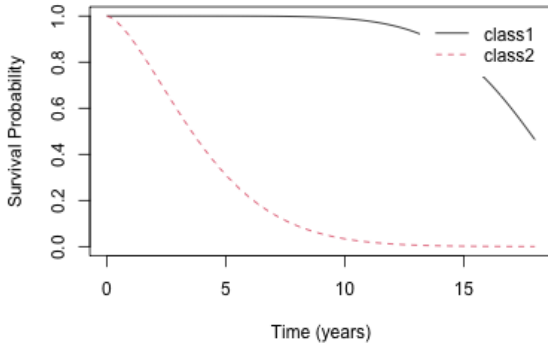
Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-1060.314	2158.243	1.000	100.00	NA	NA
Model2	2	-1032.385	2127.461	0.688	76.33	23.67	NA
Model3	3	-1031.042	2149.852	0.517	1.89	24.62	73.48

Analysis of the log-likelihood, BIC and entropy provided in **table??** suggests that the two class model in this instance provides the best method for discriminating latent classes given longitudinal NfL measurements and time-to-diagnosis data. The log-likelihood increases most between the one class and the two class model, and only increases very slightly between the two class and the three class model. The BIC is minimised and entropy is maxmised for the two class model. The posterior probabilities are also analysed in order to ensure good class discrimination.

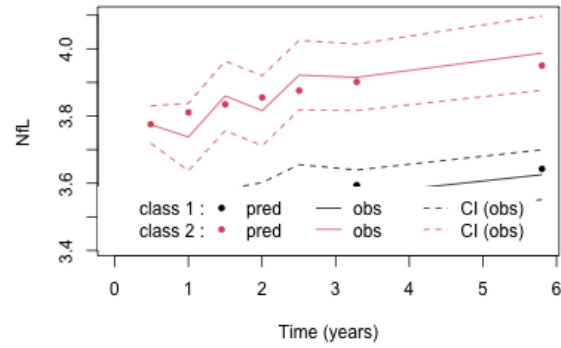
Table 9: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal NfL measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8982	0.1018	93.55	82.38	60.3
Class2	0.0270	0.9730	96.00	95.20	92.8

Latent class sizes for the two class model are reasonable, and good class discrimination is shown when assessing average posterior probabilities (**table ??**). Again, the one class model had a significant score test statistic (33.409, $p < 0.001$) which became non-significant for the two class model (0.368, $p = 0.544$). Therefore, the two class model was chosen due to support for two latent classes in the sample population.



(a) **Figure 6:** class-specific survival curves for the two class JLCMM for longitudinal NfL measurements



(a) **Figure 7:** class-specific NfL trajectories for the two class JLCMM

Figure ?? shows the class-specific survival given from the two class JLCMM with longitudinal NfL measurements. The plot shows similar survival curves to those in **figure ??**, suggesting both models have discerned latent classes with similar time-to-diagnosis processes.

Multivariate JLCMM of longitudinal p-tau181 and NfL measurements and time-

to-dementia data

The p-tau181 and NfL measurements were also modeled simultaneously with a multivariate JLCMM. Models with one, two and three classes were fit with a simple random effects structure as although the BIC was reduced with a random intercept and slope, the difference was minimal at ~ 1.1 unit decrease. All model parameters were significant ($p < 0.001$) in all candidate models, and are found in the **appendix (X, X, X)**. Although the log-likelihood decreased with increasing number of classes, the biggest decrease was between the one and two class models. Additionally, the BIC was minimised and the entropy was maximised for the two class model (**table ??**). Assessment of the posterior probabilities for the two class model (**table ??**) shows both reasonable class sizes and good class discrimination, supporting the choice of this model.

Table 10: Log-likelihood, BIC and entropy for JLCMMs with longitudinal ptau-191 and NfL measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-2841.631	5739.685	1.000	100.00	NA	NA
Model2	2	-2811.556	5704.609	0.687	24.24	75.76	NA
Model3	3	-2810.127	5726.829	0.520	1.89	25.00	73.11

Table 11: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal ptau-191 and NfL measurements

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8242	0.0521	0.1237	86.17	62.77	28.72
Class2	0.2388	0.7429	0.0183	65.00	45.00	15.00
Class3	0.0433	0.0072	0.9494	93.18	87.88	84.09

Sensitivity Analyses

Running same analysis was run with individuals with at least one biomarker measurement resulted in similar conclusions regarding optimal models. For the univariate JLCMM with longitudinal p-tau181 measurements, although BIC was minimised and entropy was maximised for the three class model, one class comprised of only 2.28% of the sample population (**appendix: table ??**). Additionally, the posterior classification table showed the model had poorer classification certainty (**appendix: table??**) compared with the two class model (**appendix: table??**). Comparing the two class model on the full data set and the two class model chosen in the main analysis, classification uncertainty decreased when including individuals with fewer than three longitudinal p-tau181 measurements (**table ?? vs appendix: table ??**). When modelling longitudinal NfL measurements with this data set, the BIC was minimised for the two class model and despite the three class model having the highest entropy, class2 contained only 1.14% of the sample population (**appendix: table ??**). Again, compared with the two

class model estimated on the data set including only those with more than 3 longitudinal NfL measurements, classification certainty decreased (**table ?? vs. appendix: table ??**).

The multivariate model XXX. THREE CLASS MODEL SAVED NOT TWO CLASS.

Defining the event time as the midpoint between diagnoses exams was also tested for JLCMM. These results also gave support to the two class univariate models for p-tau181 and NfL measurements (**appendix table?? & table??**). For NfL, the three class model did not converge, therefore these results are not presented. The multivariate model performed similarly to the model with the original event time (**table?? vs appendix table??**) and the two class model would be chosen also in this instance. **POSTPROB NOT REFERENCED.**

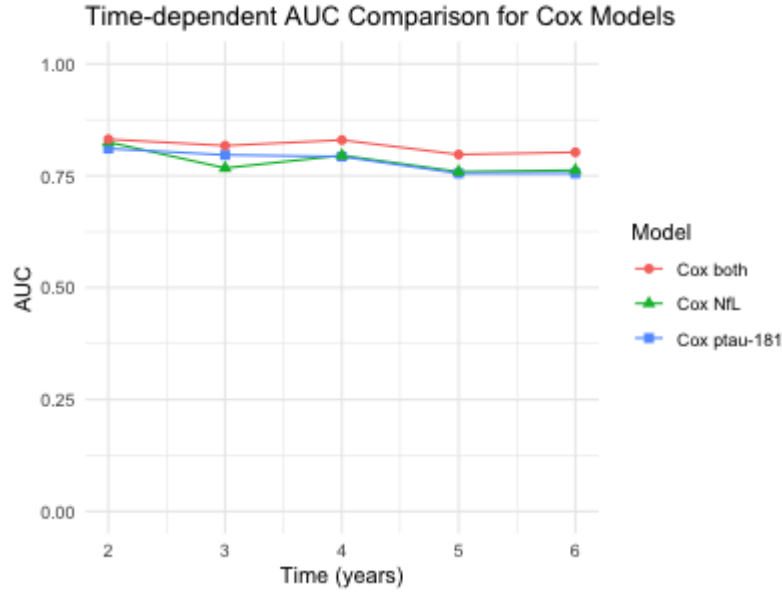
7.4 Predictive Performance

7.4.1 Time-Dependent AUC

Predictive performance was first evaluated using time-dependent AUC at yearly time points from 2 to 6 years of follow-up. Since including participants with at least one biomarker measurement did not significantly affect the precision of the model estimates, the evaluation of predictive performance was performed only for the models estimated using participants with three or more measurements. In other words, performance was not re-evaluated for the alternative specification including all participants with at least one measurement.

Figure ? provides an initial overview of the time-dependent AUC of the Cox PH models across the follow-up period.

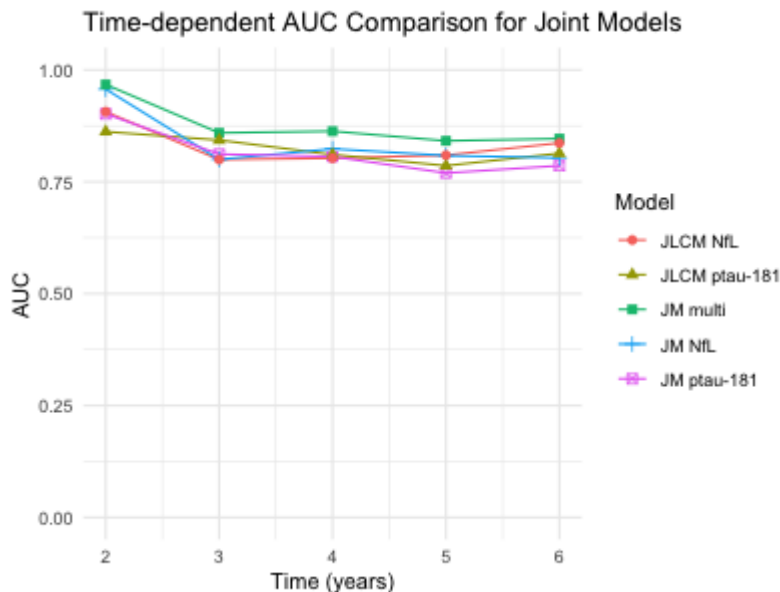
Figure ?: Time-dependent AUC curves for the Cox PH models



Overall, all Cox PH models demonstrated relatively stable predictive performance over time, with AUC values consistently approximately between 0.80 and 0.875, indicating good discriminative ability throughout the study period.

The Cox PH model including both baseline biomarker values showed the highest predictive performance across all evaluated time points. In contrast, the single-marker models demonstrated slightly lower and somewhat more variable performance over time. The trajectories of the Cox NfL and Cox p-tau181 models were broadly similar, although the NfL-only model appeared to perform slightly better during later follow-up.

Turning to the joint models and latent class joint models, Figure ? presents the time-dependent AUC across the follow-up period for these models.



Compared with the Cox PH models based only on baseline biomarker values, the joint modelling approaches generally demonstrated stronger predictive performance during earlier follow-up, with AUC values reaching up to 0.90 at the two-year time point. However, over longer follow-up, the differences between approaches became less pronounced as all models converged toward a stable, slightly lower performance range, with AUC values approximately between 0.80 and 0.87.

The joint model including multiple longitudinal biomarkers showed the strongest and most consistent predictive performance across follow-up. During the first two years of follow-up, the single-biomarker joint models also performed well. However, their performance gradually declined relative to the latent class single-biomarker joint models over time. Regardless of modelling approach, models based on longitudinal trajectories of NfL consistently performed slightly better than those based on longitudinal trajectories of p-tau181. Table ?, which presents the time-dependent AUC values for each model at each follow-up time point, confirms these findings.

7.4.2 Time-Dependent Brier Score

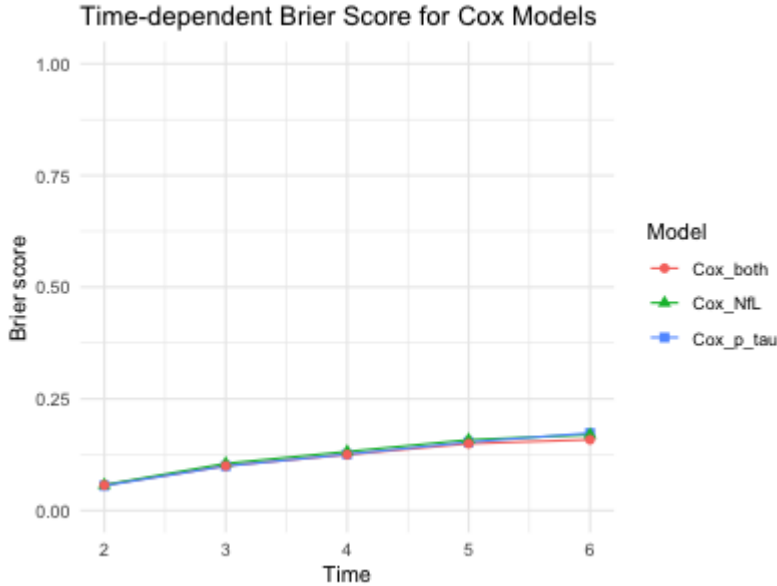
As expected, Table ? demonstrates that predictive performance is highest at two-year follow-up, particularly for joint models including multiple biomarkers. In contrast, models based on single biomarkers show lower performance, with Cox PH models based on baseline biomarkers performing least well. At later follow-up time points, differences between models are less pronounced, with AUC values converging to a narrower range between approximately 0.77 and 0.87, reflecting more similar performance across models.

Table 12: Time-dependent AUC by model and follow-up time point

Model	AUC_2y	AUC_3y	AUC_4y	AUC_5y	AUC_6y
Cox ptau-181	0.811	0.797	0.793	0.755	0.755
Cox NFL	0.825	0.767	0.795	0.759	0.762
Cox both	0.832	0.817	0.830	0.798	0.803
JM ptau-181	0.902	0.812	0.806	0.770	0.786
JM NFL	0.957	0.800	0.824	0.809	0.803
JM both	0.967	0.859	0.863	0.841	0.846
JLCM ptau-181	0.862	0.843	0.811	0.786	0.814
JLCM NFL	0.907	0.800	0.803	0.809	0.837

Additionally, predictive performance was evaluated using the time-dependent Brier score, with Figure ? providing an initial overview of the Brier scores for the Cox PH models across the follow-up period.

Figure ?: Time-dependent Brier score curves for the Cox PH models



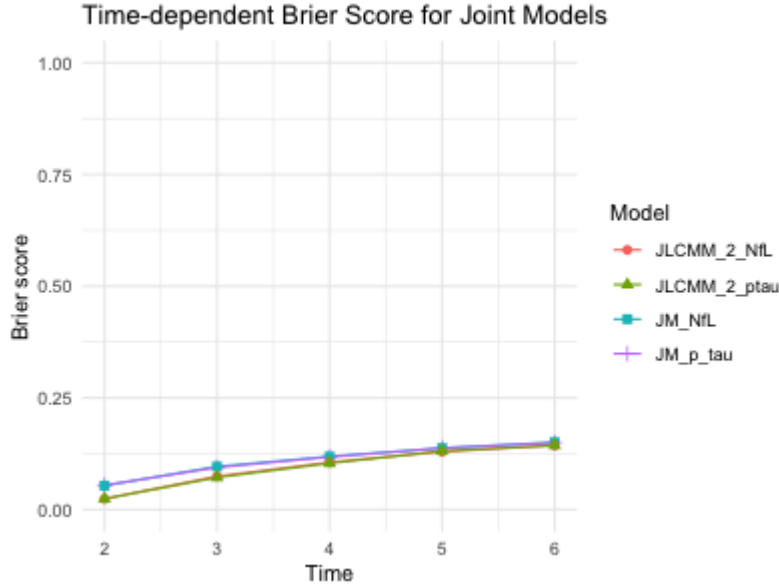
Overall, the Brier scores remain low for all Cox PH models throughout the follow-up period, indicating good overall predictive accuracy. At the two-year follow-up, all models demonstrate their lowest Brier scores, around 0.05. However, a gradual increase in Brier score is observed over time, reaching approximately 0.15 at six years, suggesting that predictions become less precise at longer follow-up times.

Differences between the Cox PH models are relatively small across the follow-up period. A

slight difference appear at the later follow up-points, where the Cox PH model including baseline values of both biomarkers shows a slightly lower Brier score than the single-biomarker models. The two single-biomarker models display very similar trajectories over time.

Figure ? presents the corresponding time-dependent Brier scores for the joint models and latent class joint models.

Figure ? : Time-dependent Brier score curves for the joint models and latent class joint models



Similar to the Cox PH models, the Brier scores are lowest at earlier follow-up times and increase gradually over time. At the two-year follow-up, however, the latent class joint models show slightly lower Brier scores than the other joint models and the Cox PH models. Differences between the joint models are otherwise small and become less pronounced over follow-up time, with Brier scores converging to around 0.15 across models at six years.

The same pattern can be observed in Table ?, which presents the time-dependent Brier score values for each model at each follow-up time point.

Table 13: Time-dependent Brier score by model and follow-up time point

Time	Cox_p_tau	Cox_NfL	Cox_both	JM_p_tau	JM_NfL	JLCMM_2_NfL	JLCMM_2_ptau
2	0.0558451	0.0573431	0.0568420	0.0544988	0.0534229	0.0239109	0.0241973
3	0.0995319	0.1052166	0.0999645	0.0941693	0.0963788	0.0744236	0.0718949
4	0.1259382	0.1318881	0.1252026	0.1176499	0.1188081	0.1057930	0.1035886
5	0.1530793	0.1581811	0.1498140	0.1376247	0.1376509	0.1294139	0.1311481
6	0.1734591	0.1696399	0.1586728	0.1493872	0.1506884	0.1436158	0.1435247

Table ? confirms that at the two-year follow-up, all Cox and joint models have low Brier scores, with the latent class joint models showing the lowest values. Brier scores increase gradually over time for all models, and by six years they converge around 0.14-0.16, with only minor differences between models. The Cox PH model including both baseline biomarkers and the joint models generally perform slightly better than the single-biomarker Cox PH models, confirming the patterns observed in the figures.

8 Discussion

8.1 Main Findings/Summary

Summarise the results.

8.2 Conclusions

Interpret the results. What did we find?

8.3 Discussion - Relating to the research question(s)/previous literature

How do our results help to answer the research question? Anything unexpected/not in accordance with previous studies?

Kanske problem vi stötte på?

8.4 Further Research

Suggest what kinds of questions this analysis leaves us with? What more can be done/looked at?

Since the exact dates of Alzheimer's disease onset are not available in the dataset and only diagnostic status at each visit is recorded, the event times for progression to dementia were consequently approximated based on observed follow-up visits. In other words, participants were treated as right-censored at the time of their last known assessment, rather than as interval-censored. However, it is highly likely that disease progression occurred between visits rather than exactly at the recorded observation times, which introduces a potential bias in the estimated event times.

Despite this, interval-censoring was not implemented given the level of this thesis and its time constraints. Properly accounting for interval-censored event times would require more complex statistical methods, for which standard Cox PH models would no longer be directly applicable.

As a direction for future research, it is therefore recommended that progression to dementia be modeled using interval-censored survival methods in order to reduce potential bias and better reflect the underlying disease process.

Furthermore, as plasma biomarker research is still emerging and longitudinal data with many repeated measurements per individual remain limited, future studies should aim to replicate and extend this analysis once larger cohorts with more frequent longitudinal sampling become available. This would improve the precision of both trajectory estimation and the assessment of associations between biomarker dynamics and disease progression.

9 Bibliography

Alzheimer's Society (n.d.). *Risk factors for Alzheimer's disease*. <https://www.alzheimers.org.uk/about-dementia/types-dementia/who-gets-alzheimers-disease>

Alzheimer's Disease Neuroimaging Initiative (ADNI) (n.d.-a). *About ADNI*. <https://adni.loni.usc.edu/about/>

- (n.d.-b). *ADNI Data*. <https://adni.loni.usc.edu/data-samples/adni-data/>

Alzheimer's Association (n.d.). Mild cognitive impairment (MCI). https://www.alz.org/alzheimers-dementia/what-is-dementia/related_conditions/mild-cognitive-impairment

Biomarkers Definitions Working Group. (2001). Biomarkers and surrogate endpoints: preferred definitions and conceptual framework". *Clinical pharmacology and therapeutics*, 69(3), 89-95. <https://doi.org/10.1067/mcp.2001.113989>

Brown, E. R., & Ibrahim, J. G. (2003). A Bayesian semiparametric joint hierarchical model for longitudinal and survival data. *Biometrics*, 59(2), 221-228. <https://doi.org/10.1111/1541-0420.00028>

Carlin, B. P., & Louis, T. A. (2008). *Bayesian methods for data analysis* (3rd ed.). CRC Press.

Carrillo, M. C., Dean, R. A., Nicolas, F., Miller, D. S., Berman, R., Khachaturian, Z., Bain, L. J., Schindler, R., Knopman, D., & Alzheimer's Association Research Roundtable (2013). Revisiting the framework of the National Institute on Aging-Alzheimer's Association diagnostic criteria. *Alzheimer's & dementia : the journal of the Alzheimer's Association*, 9(5), 594-601. <https://doi.org/10.1016/j.jalz.2013.05.1762>

Celeux, G. & Soromenho, G. (1996). An entropy criterion for assessing the number of clusters in a mixture model. *Journal of Classification*: 13(1), 195-212. <https://doi.org/10.1007/BF01246098>

Chang, L., Rissin, D. M., Fournier, D. R., Piech, T., Patel, P. P., Wilson, D. H., & Duffy, D. C. (2012). Single Molecule Enzyme-Linked Immunosorbent Assays:

Theoretical Considerations. *Journal of Immunological Methods*, 378(1-2), 102–115. <https://doi.org/10.1016/j.jim.2012.02.011>

Chen, S. D., Huang, Y. Y., Shen, X. N., Guo, Y., Tan, L., Dong, Q., Yu, J. T., & Alzheimer’s Disease Neuroimaging Initiative (2021). Longitudinal plasma phosphorylated tau 181 tracks disease progression in Alzheimer’s disease. *Translational psychiatry*, 11(1), 356. <https://doi.org/10.1038/s41398-021-01476-7>

Clark, C., Piotr Lewczuk, Johannes Kornhuber, Richiardi, J., Bénédicte Maréchal, Karikari, T. K., Kaj Blennow, Zetterberg, H., & Popp, J. (2021). Plasma neurofilament light and phosphorylated tau 181 as biomarkers of Alzheimer’s disease pathology and clinical disease progression. *Alzheimer’s Research & Therapy*, 13(1). <https://doi.org/10.1186/s13195-021-00805-8>

Fitzmaurice, G. M., Laird, N. M., & Ware, J. H. (2011). *Applied longitudinal analysis* (2nd ed.). John Wiley & Sons.

Guo, G., Song, W., Wang, A., Cui, Q., Yang, X., Wang, Y., Ma, Y., Han, H., Li, Z., Zhang, Z., Meng, W., & Wang, S. (2026). Sensitivity comparison of longitudinal cognitive function indicators of Alzheimer’s disease after mild cognitive impairment: a prospective cohort study. *Sci Rep*. <https://doi.org/10.1038/s41598-026-44192-2>

Hamra, G., MacLehose, R., & Richardson, D. (2013). Markov chain Monte Carlo: an introduction for epidemiologists. *International journal of epidemiology*, 42(2), 627–634. <https://doi.org/10.1093/ije/dyt043>

Jack, C. R., Andrews, J. S., Beach, T. G., Buracchio, T., Dunn, B., Graf, A., Hansson, O., Ho, C., Jagust, W., McDade, E., Molinuevo, J. L., Okonkwo, O. C., Pani, L., Rafii, M. S., Scheltens, P., Siemers, E., Snyder, H. M., Sperling, R., Teunissen, C. E., & Carrillo, M. C. (2024). Revised criteria for diagnosis and staging of Alzheimer’s disease: Alzheimer’s Association Workgroup. *Alzheimer’s & Dementia*, 20(8). <https://doi.org/10.1002/alz.13859>

Jacqmin-Gadda, H., Proust-Lima, C., Taylor, J. M. G., & Commenges, D. (2010). Score Test for Conditional Independence Between Longitudinal Outcome and Time to Event Given the Classes in the Joint Latent Class Model. *Biometrics*, 66(1), 11–19. <http://www.jstor.org/stable/40663147>

Janelidze, S., Mattsson, N., Palmqvist, S., Smith, R., Beach, T. G., Serrano, G. E., Chai, X., Proctor, N. K., Eichenlaub, U., Zetterberg, H., Blennow, K., Reiman, E. M., Stomrud, E., Dage, J. L., & Hansson, O. (2020). Plasma P-tau181 in Alzheimer’s disease: relationship to other biomarkers, differential diagnosis, neuropathology and longitudinal progression to Alzheimer’s dementia. *Nature medicine*, 26(3), 379–386. <https://doi.org/10.1038/s41591-020-0755-1>

Kamarudin, A. N., Cox, T., & Kolamunnage-Dona, R. (2017). Time-dependent ROC curve analysis in medical research: Current methods and applications. *BMC Medical Research Methodology*, 17, 53. <https://doi.org/10.1186/s12874-017-0332-6>

Kleinbaum, D. G., & Klein, M. (2012). *Survival analysis: A self-learning text* (3rd ed.). Springer.

Kyheng, M., Babykina, G., & Duhamel, A. (2025). Joint Latent Class Models: A Tutorial on Practical Applications in Clinical Research. *Stat Med*, 44(8-9):e70047. 10.1002/sim.70047. PMID: 40277369; PMCID: PMC12023844.

Mattsson, N., Andreasson, U., Zetterberg, H., & Blennow, K. (2017). Association of Plasma Neurofilament Light With Neurodegeneration in Patients With Alzheimer Disease. *JAMA Neurology*, 74(5), 557. <https://doi.org/10.1001/jamaneurol.2016.6117>

Mattsson, N., Cullen, N. C., Andreasson, U., Zetterberg, H., & Blennow, K. (2019). Association Between Longitudinal Plasma Neurofilament Light and Neurodegeneration in Patients With Alzheimer Disease. *JAMA neurology*, 76(7), 791-799. <https://doi.org/10.1001/jamaneurol.2019.0765>

Moore, D. F. (2016). Regression Analysis Using the Proportional Hazards Model. *Use R!*, 55–72. https://doi.org/10.1007/978-3-319-31245-3_5

Peng, R.E. (2022). *Advanced Statistical Computing*. <https://bookdown.org/rdpeng/advstatcomp/monitoring-convergence.html#gelman-rubin-statistic>

Proust-Lima, C., Saulnier, T., Philipps, V., Traon, A. P., Péran, P., Rascol, O., Meissner, W. G., & Foubert-Samier, A. (2023). Describing complex disease progression using joint latent class models for multivariate longitudinal markers and clinical endpoints. *Statistics in medicine*, 42(22), 3996-4014. <https://doi.org/10.1002/sim.9844>

Rissin, D. M., Kan, C. W., Campbell, T. G., Howes, S. C., Fournier, D. R., Song, L., Piech, T., Patel, P. P., Chang, L., Rivnak, A. J., Ferrell, E. P., Randall, J. D., Provuncher, G. K., Walt, D. R., & Duffy, D. C. (2010). Single-molecule enzyme-linked immunosorbent assay detects serum proteins at subfemtomolar concentrations. *Nature Biotechnology*, 28(6), 595–599. <https://doi.org/10.1038/nbt.1641>

Rizopoulos, D. (2012). *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R* (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/b12208>

- (2018). *Multivariate joint models*.

Scollard, P., Mukherjee, S., Choi, S. E., Lee, M. L., Klinedinst, B., Gibbons, L. E., Trittschuh, E. H., Mez, J., Saykin, A. J., James, B. D., Proust-Lima, C., & Crane, P. K. (2026). Heterogeneous patterns of cognitive decline in Alzheimer’s disease across three domains of cognition. *Journal of Alzheimer’s disease : JAD*, 110(1), 383-396. <https://doi.org/10.1177/13872877251414975>

Seo Young Park, Ji Eun Park, Kim, H., & Seong Ho Park. (2021). Review of Statistical Methods for Evaluating the Performance of Survival or Other Time-to-Event Prediction Models (from Conventional to Deep Learning Approaches). *Korean Journal of Radiology*, 22(10), 1697–1697. <https://doi.org/10.3348/kjr.2021.0223>

Therneau, T.M. & Grambsch, P.M (2000). *Modeling Survival Data: Extending the Cox Model*. Springer, Berlin. <https://doi.org/10.1007/978-1-4757-3294-8>

Verbeke, G., & Geert Molenberghs. (2000). *Linear Mixed Models for Longitudinal Data*. Springer Science & Business Media.

Wang, C., Shen, J., Charalambous, C., & Pan, J. (2024). Modeling biomarker variability in joint analysis of longitudinal and time-to-event data. *Biostatistics* (Oxford, England), 25(2), 577-596. <https://doi.org/10.1093/biostatistics/kxad009>

Wong, P.C., Savonenko, A., Li, T., Price, DL. (2012). ‘Neurobiology of Alzheimer’s Disease’ in Brady, S.T., Siegel, G.J., Albers, RW., Price, D.L (8 ed.) *Basic neurochemistry : principles of molecular, cellular, and medical neurobiology*. Waltham, Massachusetts; Oxford: Academic Press / Elsevier.

World Health Organization (2024). *Preferred product characteristics of blood-based biomarker diagnostics for Alzheimer disease*. <https://iris.who.int/server/api/core/bitstreams/43c83ef7-7a34-4b56-ae17-b584f2fc0798/content>

- (2025). *Dementia*. <https://www.who.int/news-room/fact-sheets/detal/dementia>

Wu, L., Liu, W., & Hu, X. J. (2010). Joint inference on HIV viral dynamics and immune suppression in presence of measurement errors. *Biometrics*, 66(2), 327-335. <https://doi.org/10.1111/j.1541-0420.2009.01308.x>

Xia, X., Duffner, L. A., Bintener, C., Bradshaw, A., Lamirel, D., & Jönsson, L. (2025). Diagnostic and prognostic multimodal prediction models in Alzheimer’s disease: A scoping review. *Journal of Alzheimer’s disease : JAD*, 108(1), 209-221. <https://doi.org/10.1177/13872877251351630>

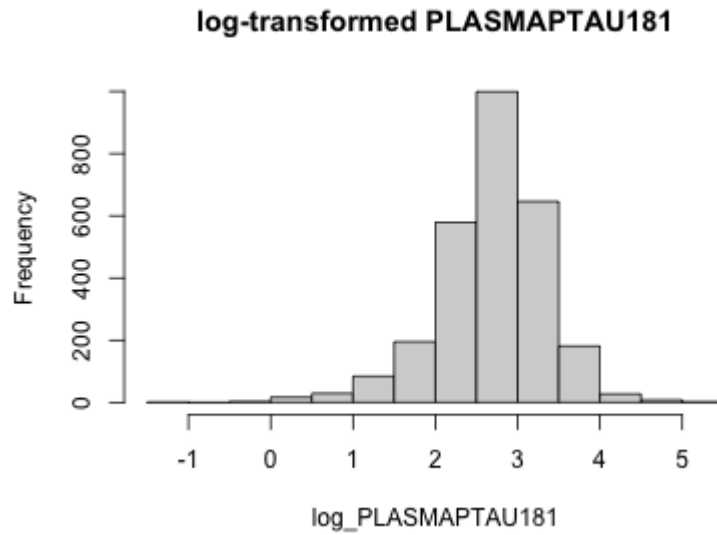
Xu, L. (2021). Bayesian Multivariate Joint Modeling for Skewed-longitudinal and Time-to-event Data. *USF Tampa Graduate Theses and Dissertations*. <https://digitalcommons.usf.edu/etd/9268>

Yuan, M., Lian, S., Li, X., Long, X., Fang, Y., & Alzheimer’s Disease Neuroimaging Initiative (ADNI) (2024). Blood biomarkers in dynamic prediction of conversion to Alzheimer’s disease: An application of joint modeling. *International journal of geriatric psychiatry*, 39(3), e6079. <https://doi.org/10.1002/gps.6079>

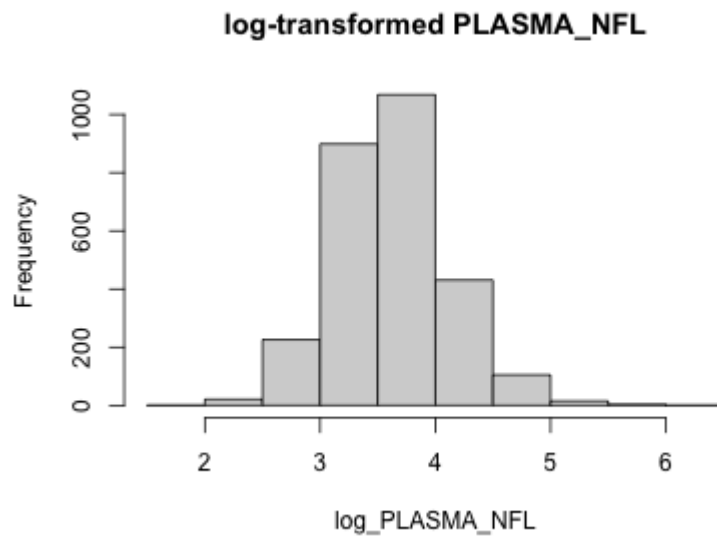
Zou, K. H., Liu, A., Bandos, A. I., Ohno-Machado, L., & Rockette, H. E. (2012). Statistical evaluation of diagnostic performance: Topics in ROC analysis (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/b11031>

10 Appendix

10.1 Appendix X: distribution of p-tau181 after log-transformation



10.2 Appendix X: distribution of NfL after log-transformation



10.3 Appendix X: correlation plot over the numerical variables used in the Cox PH models

10.4 Appendix X: correlation plot over the numerical variables used in the LME models

10.5 Appendix X

Table 14: Results of the likelihood ratio tests to determine the random effects structure for LME models

Model	Test statistic	P-value
ptau-181	1.488	0.475
NfL	10.506	0.005

10.6 Appendix X

Table 15: P-values for the proportional hazards assumption tests for all covariates in the Cox PH models

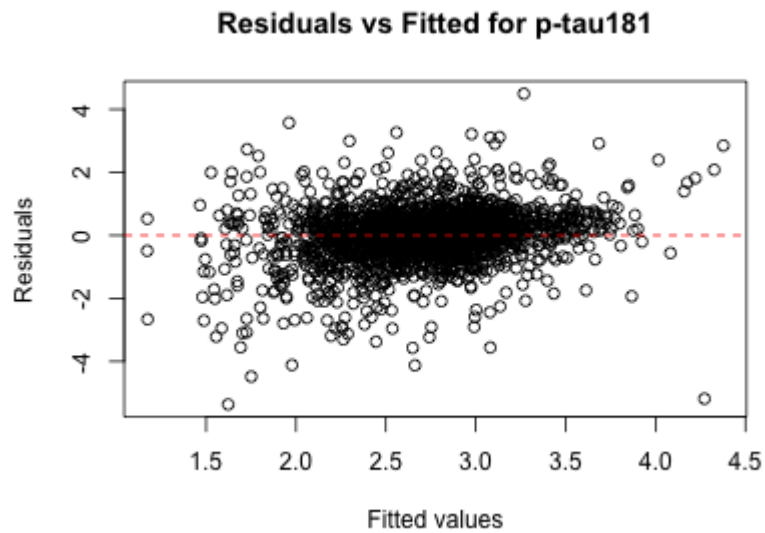
variable	Standard	p-tau181 baseline	NfL baseline	Both baseline
age_at_baseline	0.002	0.002	0.003	0.002
PTGENDER	0.803	0.835	0.827	0.888
APOE4_status	0.158	0.173	0.127	0.154
GLOBAL	0.013	0.028	0.031	0.057
baseline_value_ptau	NA	0.698	NA	0.632
baseline_value_nfl	NA	NA	0.140	0.113

10.7 Appendix X

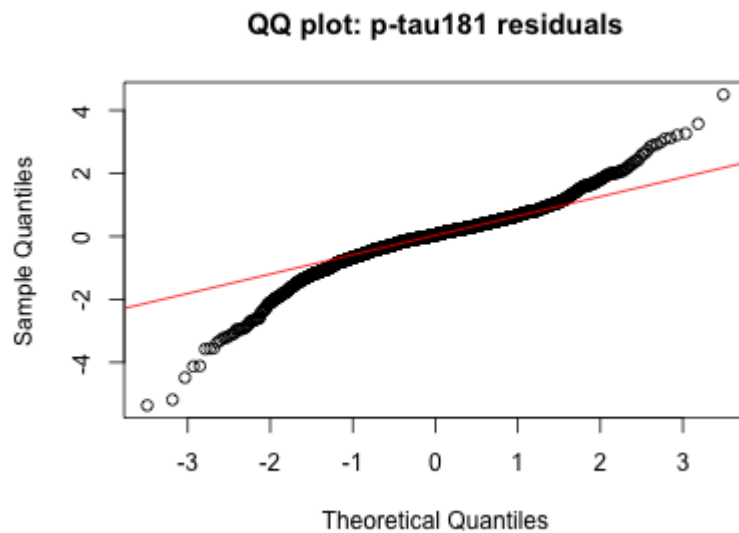
Table 16: Results of the likelihood ratio test for the Cox PH models with and without gender as a covariate

Model	Test statistic	P-value
ptau-181 baseline	0.3824835	0.98
NfL baseline	1.0508101	0.95
ptau-181 and NfL baseline	0.5706900	0.73

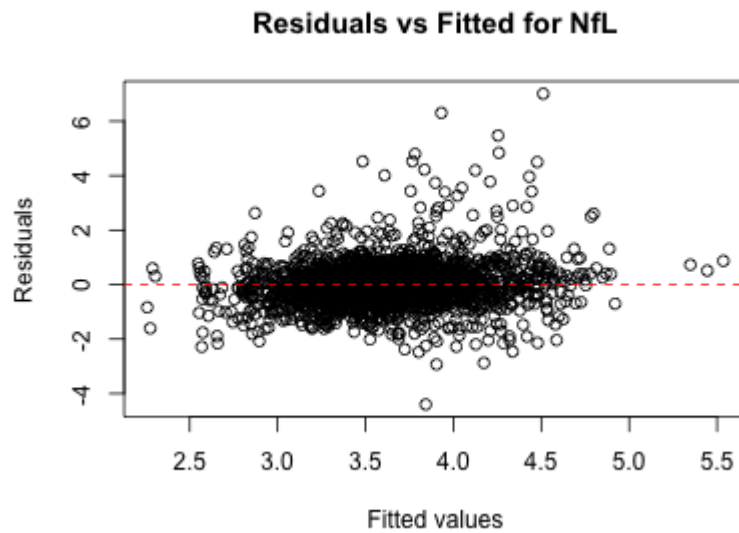
10.8 Appendix X: fitted values versus residuals from the LME model for p-tau181



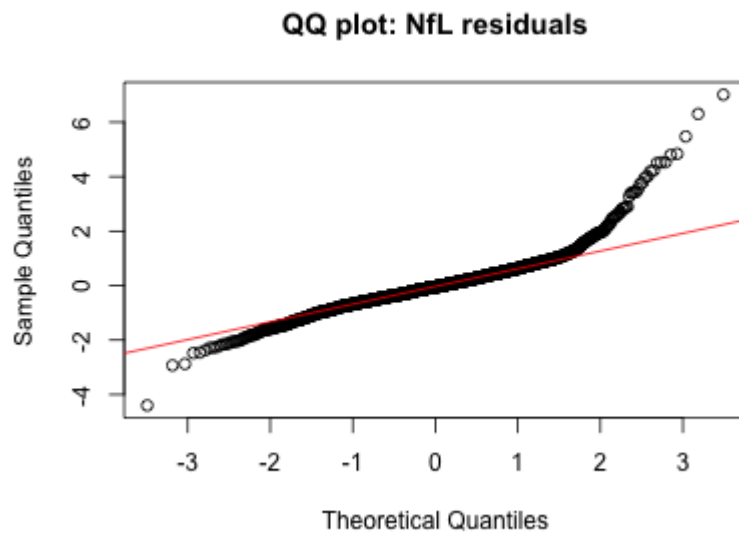
10.9 Appendix X: Q-Q plot of the LME model for p-tau181



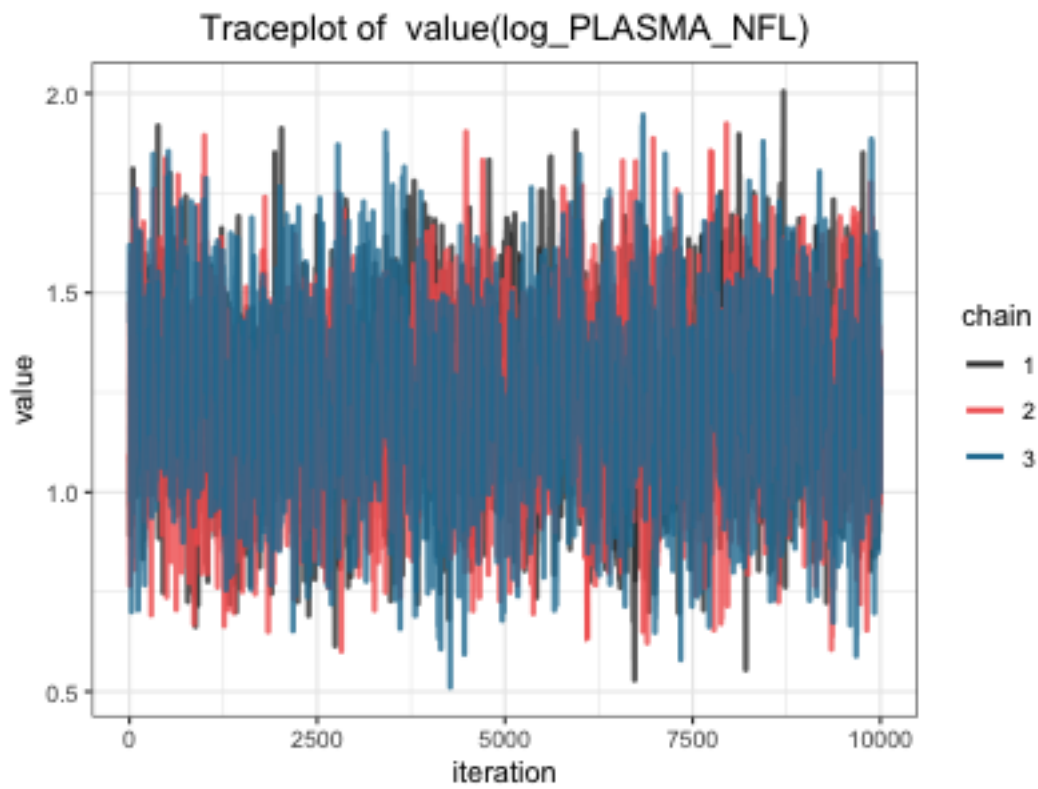
10.10 Appendix X: fitted values versus residuals from the LME model for NfL



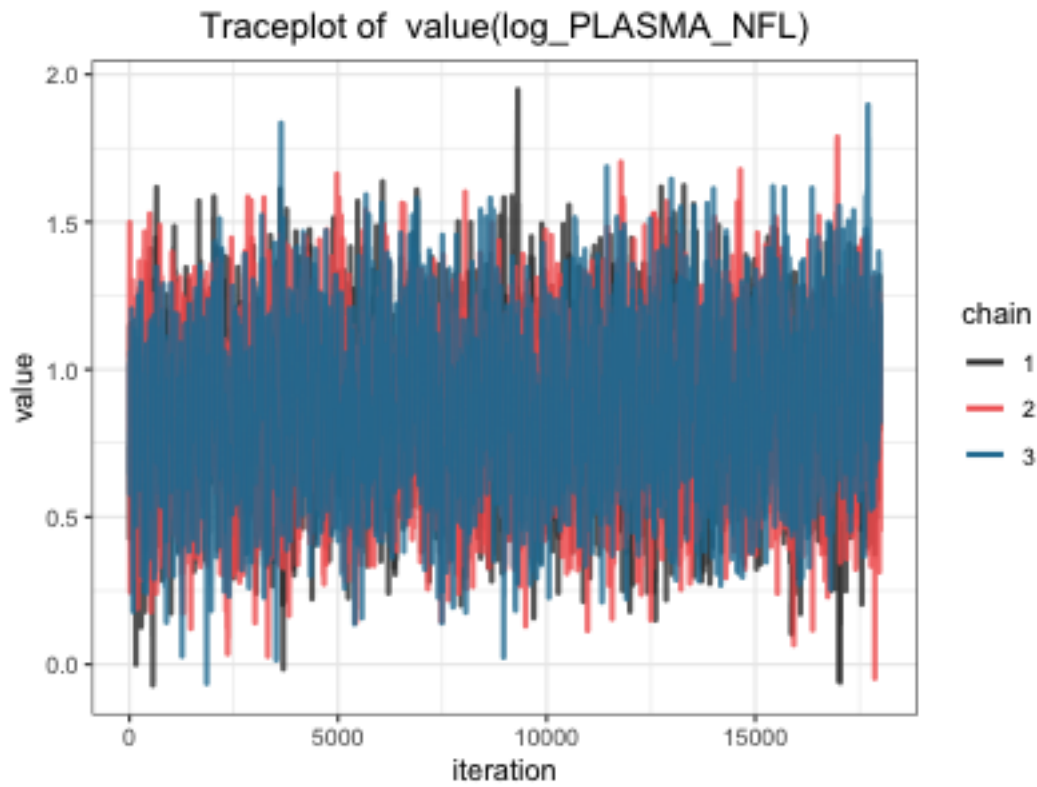
10.11 Appendix X: Q-Q plot of the LME model for NfL



10.12 Appendix X: trace plot of the α -parameter for the joint model of NFL



10.13 Appendix X: trace plot of the α -parameter for the multivariate joint model



10.14 Sensitivity Analysis: 1+ Biomarker Measurements

Table 17: Summary of fit statistics for JLCMM with 1+ longitudinal measurements for p-tau181

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)	Class4 (%)
Model1	1	-2432.947	4905.208	1.000	100.00	NA	NA	NA
Model2	2	-2390.094	4845.714	0.585	23.82	76.18	NA	NA
Model3	3	-2363.471	4818.676	0.711	2.28	72.61	25.11	NA
Model4	4	-2361.451	4840.848	0.650	2.28	1.85	24.25	71.61

TABLE IS SAVED WRONG -> IS THE THREE CLASS MODEL

10.15 Sensitivity Analysis: Interval Censored Events

Table 18: Summary of posterior probabilities for the two class JLCMM with 1+ longitudinal measurements for p-tau181

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9218	0.0782	85.03	81.44	80.24
Class2	0.1393	0.8607	86.33	69.48	47.94

Table 19: Summary of posterior probabilities for the three class JLCMM with 1+ longitudinal measurements for p-tau181

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8279	0.1703	0.0018	75.00	56.25	56.25
Class2	0.0098	0.8424	0.1478	82.32	66.01	43.42
Class3	0.0000	0.0829	0.9171	85.80	80.68	75.57

Table 20: Summary of fit statistics for JLCMM with 1+ longitudinal measurements for NFL

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-2432.947	4905.208	1.000	100.00	NA	NA
Model2	2	-1268.803	2603.131	0.651	18.54	81.46	NA
Model3	3	-1263.410	2618.556	0.748	76.32	1.14	22.54

Table 21: Summary of posterior probabilities for the two class JLCMM with 1+ longitudinal measurements for NFL

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8136	0.1864	75.38	59.23	43.08
Class2	0.0810	0.9190	90.89	85.46	73.73

Table 22: Summary of fit statistics for JLCMM with 1+ longitudinal measurements for p-tau181 and NFL

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-3521.122	7101.216	1.000	100.00	NA	NA
Model2	2	-3467.613	7020.408	0.644	75.46	24.54	NA
Model3	3	-3465.765	7042.922	0.576	1.14	76.18	22.68

Table 23: Summary of posterior probabilities for the two class JLCMM with 1+ longitudinal measurements for p-tau181 and NfL

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8279	0.1703	0.0018	75.00	56.25	56.25
Class2	0.0098	0.8424	0.1478	82.32	66.01	43.42
Class3	0.0000	0.0829	0.9171	85.80	80.68	75.57

Table 24: P-values for the proportional hazards assumption tests after exclusion of age at baseline in the Cox PH models

variable	Standard	p-tau181 baseline	NfL baseline	Both baseline
PTGENDER	0.758	0.835	0.821	0.911
APOE4_status	0.161	0.170	0.126	0.148
GLOBAL	0.293	0.430	0.136	0.232
baseline_value_ptau	NA	0.745	NA	0.658
baseline_value_nfl	NA	NA	0.105	0.085

Table 25: Hazard ratios with 95% confidence intervals and p-values from the Cox PH models including baseline biomarkers

Variable	PTau HR (CI)	PTau P	NfL HR (CI)	NfL P	Both HR (CI)	Both P
PTGENDERMale	0.985 (0.694–1.396)	0.931	0.98 (0.69–1.392)	0.912	0.933 (0.656–1.328)	0.702
APOE4_status1 allele	2.04 (1.404–2.966)	<0.001	2.215 (1.525–3.217)	<0.001	2.048 (1.408–2.978)	<0.001
APOE4_status2 alleles	3.626 (2.1–6.261)	<0.001	4.451 (2.585–7.665)	<0.001	3.846 (2.224–6.653)	<0.001
baseline_value_ptau	1.991 (1.485–2.67)	<0.001	NA (NA–NA)	NA	1.749 (1.282–2.386)	<0.001
baseline_value_nfl	NA (NA–NA)	NA	2.374 (1.632–3.453)	<0.001	1.968 (1.331–2.911)	<0.001

Table 26: Concordance indices for all fitted Cox PH models with baseline biomarker values

Model	C-Index
Cox ptau-181	0.679
Cox NfL	0.673
Cox ptau-181 & NfL	0.681

Table 27: $\widehat{\mathbf{R}}$ values of the survival outcome in the joint models

Variable	PTau	NfL	Multi
PTGENDERMale	1.002	1.000	1.000
APOE4_status1 allele	1.001	1.004	1.001
APOE4_status2 alleles	1.001	1.001	1.000
value(log_PLASMAPTAU181)	1.009	NA	1.002
value(log_PLASMA_NFL)	NA	1.008	1.001

Table 28: $\widehat{\mathbf{R}}$ values of the longitudinal outcome in the joint models

Variable	Long_PTau	Long_NfL	Long_Multi
(Intercept)	1.001	1.001	1.001
time_years	1.022	1.020	1.036
PTGENDERMale	1.000	1.001	1.001
APOE4_status1 allele	1.001	1.001	1.000
APOE4_status2 alleles	1.000	1.000	1.000
sigma	1.005	1.032	1.016

Table 29: Hazard ratios with 95% confidence intervals and p-values from the joint models

Variable	PTau HR (CI)	PTau P	NfL HR (CI)	NfL P	Both HR (CI)	Both P
PTGENDERMale	0.917 (0.581–1.449)	0.72	0.963 (0.615–1.523)	0.874	0.881 (0.556–1.401)	0.61
APOE4_status1 allele	1.902 (1.252–2.842)	0.001	2.294 (1.566–3.408)	<0.001	2.018 (1.347–3.033)	<0.001
APOE4_status2 alleles	3.196 (1.798–5.505)	<0.001	4.548 (2.562–7.803)	<0.001	3.704 (2.056–6.499)	<0.001
value(log_PLASMAPTAU181)	3.241 (2.135–4.887)	<0.001	NA (NA–NA)	NA	2.229 (1.381–3.664)	<0.001
value(log_PLASMA_NFL)	NA (NA–NA)	NA	3.332 (2.218–4.995)	<0.001	2.345 (1.475–3.749)	<0.001

Table 30: Table 6: log-likelihood, BIC and entropy for JLCMMs with longitudinal p-tau181 measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)	Class4 (%)
Model1	1	-2013.895	4065.405	1.000	100.00	NA	NA	NA
Model2	2	-1987.444	4037.579	0.615	24.81	75.19	NA	NA
Model3	3	-1982.269	4052.305	0.637	26.89	3.98	69.13	NA
Model4	4	-1970.862	4054.568	0.721	21.78	7.20	67.80	3.22

Table 31: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal p-tau181 measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9460	0.0540	93.13	87.79	85.50
Class2	0.1224	0.8776	92.19	77.33	50.88

Table 32: Posterior classification table and mean posterior probabilities for the three class JLCMM for longitudinal p-tau181 measurements

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9265	0.0093	0.0642	88.73	82.39	78.87
Class2	0.0180	0.7348	0.2473	61.90	42.86	14.29
Class3	0.1403	0.0476	0.8121	83.84	59.45	24.66

Table 33: Log-likelihood, BIC and entropy for JLCMMs with longitudinal NFL measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)
Model1	1	-2013.895	4065.405	1.000	100.0	NA
Model2	2	-1030.263	2123.217	0.704	23.3	76.7

Table 34: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal NFL measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9691	0.0309	95.12	94.31	91.06
Class2	0.0880	0.9120	95.80	86.42	64.20

Table 35: Log-likelihood, BIC and entropy for JLCMMs with longitudinal p-tau181 and NFL measurements

Candidate model	G	Log Likelihood	BIC	Entropy	Class1 (%)	Class2 (%)	Class3 (%)
Model1	1	-2835.832	5728.086	1.000	100.00	NA	NA
Model2	2	-2809.520	5700.538	0.699	23.67	76.33	NA
Model3	3	-2808.078	5722.730	0.633	1.89	24.24	73.86

Table 36: Posterior classification table and mean posterior probabilities for the two class JLCMM for longitudinal p-tau181 and NFL measurements

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9643	0.0357	96.00	92.8	89.60
Class2	0.0897	0.9103	95.53	86.6	63.03